

LAPORAN TUGAS PRAKTIKUM 1

Mata Kuliah Sistem Operasi



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12 Oktober 2025**

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1 ISI LAPORAN

1.1 Operating – System Services

1.1.1 Menjelajahi layanan OS

- Menampilkan struktur perangkat penyimpanan (block devices) di sistem

```
user_rafi@DESKTOP-AC7MIM X + v
user_rafi@DESKTOP-AC7MIM2:~$ uname -a
Linux DESKTOP-AC7MIM2 6.6.87.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu Jun 5 18:30:46 UTC 2025 x86_64 x86_64 x86_64 GNU/Linux
user_rafi@DESKTOP-AC7MIM2:~$ lsb_release -a
No LSB modules are available.
Distributor ID: Ubuntu
Description: Ubuntu 24.04.3 LTS
Release: 24.04
Codename: noble
user_rafi@DESKTOP-AC7MIM2:~$ ps aux | head -20
USER      PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root         1   0.0  0.0      0     0 ttyS0    Ss+  18:47   0:02 /sbin/init
root         2   0.0  0.0      0     0 ttyS0    Sl   18:47   0:00 /init
root         7   0.0  0.0      0     0 ttyS0    Sl   18:47   0:00 plan9 --control-socket 7 --log-level 4 --server-fd 8 --pipe-fd 10 --log-truncate
root        49   0.3  0.3 58640 15376 ?        S<s  18:47   0:00 /usr/lib/systemd/systemd-journald
root       102   0.1  0.1 25268  6272 ?        Ss   18:47   0:00 /usr/lib/systemd/systemd-udev
systemd+  117   0.1  0.3 21456 12416 ?        Ss   18:47   0:00 /usr/lib/systemd/systemd-resolved
systemd+  118   0.0  0.1 91024  7552 ?        Ssl  18:47   0:00 /usr/lib/systemd/systemd-timesyncd
root       168   0.0  0.0  4236  2560 ?        Ss   18:47   0:00 /usr/sbin/cron -f -p
message+  169   0.0  0.1  9596  5120 ?        Ss   18:47   0:00 @dbus-daemon --system --address=systemd: --nofork --nopidfile --systemd-activation --syslog-only
root       176   0.0  0.2 17960  8576 ?        Ss   18:47   0:00 /usr/lib/systemd/systemd-logind
root       178   0.0  0.3 1756096 12800 ?       Ssl  18:47   0:00 /usr/libexec/wnsl-pro-service -vv
root       180   0.0  0.0  3160  2048 hvc0    Ss+  18:47   0:00 /sbin/agetty -o -p -- \u --noclear --keep-baud - 115200,38400,9600 vt220
root       184   0.0  0.0  3116  1920 tty1    Ss+  18:47   0:00 /sbin/agetty -o -p -- \u --noclear - linux
syslog    185   0.0  0.1 222508  5504 ?        Ssl  18:47   0:00 /usr/sbin/rsyslogd -n -iNONE
root       193   0.2  0.5 107012 22400 ?       Ssl  18:47   0:00 /usr/bin/python3 /usr/share/unattended-upgrades/unattended-upgrade-shutdown --wait-for-signal
root       300   0.0  0.0  3076    896 ?        Ss   18:47   0:00 /init
root       301   0.0  0.0  3092   1024 ?        S   18:47   0:00 /init
user_ra+  302   0.0  0.1  6072   5120 pts/0    Ss   18:47   0:00 -bash
root       303   0.0  0.1  6664  4480 pts/1    Ss   18:47   0:00 /bin/login -f
user_rafi@DESKTOP-AC7MIM2:~$ free -h
              total        used        free      shared  buff/cache   available
Mem:           3.7Gi        363Mi       3.0Gi         3.4Mi        418Mi       3.3Gi
Swap:          1.0Gi           0B         1.0Gi
```

```
user_rafi@DESKTOP-AC7MIM X + v
user_rafi@DESKTOP-AC7MIM2:~$ lscpu
Architecture:          x86_64
CPU op-mode(s):        32-bit, 64-bit
Address sizes:          39 bits physical, 48 bits virtual
Byte Order:             Little Endian
CPU(s):                 4
On-line CPU(s) list:   0-3
Vendor ID:              GenuineIntel
Model name:             11th Gen Intel(R) Core(TM) i3-1115G4 @ 3.00GHz
CPU family:             6
Model:                 140
Thread(s) per core:    2
Core(s) per socket:    2
Socket(s):              1
Stepping:               1
BogoMIPS:               4377.59
Flags:                  fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr sse sse2 ss ht syscall nx pdpe1gb
rdtscp lm constant_tsc arch_perfmon rep_good nopl xtopology tsc_reliable nonstop_tsc cpuid tsc_known_freq pni pclmulqdq v
mx ssse3 fma cx16 pdcm pcid sse4_1 sse4_2 x2apic movbe popcnt tsc_deadline_timer aes xsave avx f16c rdrand hypervisor lahf
_lm abm 3dnowprefetch ssbd ibrs ibpb stibp ibrs_enhanced tpr_shadow ept vpid ept_ad fsgsbase tsc_adjust bmi1 avx2 smep bmi
2 erms invpcid avx512f avx512dq rdseed adx smap avx512ifma clflushopt clwb avx512cd sha_ni avx512bw avx512vl xsaveopt xsav
ec xgetbv1 xsaves vnmix avx512vbmi umip avx512_vbmi2 gfni vaes vpclmulqdq avx512_vnni avx512_bitalg avx512_vpopcntdq rdpid
movdiri movdir64b fsrm avx512_vp2intersect md_clear flush_l1d arch_capabilities

Virtualization features:
Virtualization:         VT-x
Hypervisor vendor:      Microsoft
Virtualization type:    full
Caches (sum of all):
L1d:                    96 KiB (2 instances)
L1i:                    64 KiB (2 instances)
L2:                     2.5 MiB (2 instances)
L3:                     6 MiB (1 instance)
NUMA:
NUMA node(s):           1
NUMA node0 CPU(s):     0-3
Vulnerabilities:
Gather data sampling:   Unknown: Dependent on hypervisor status
Itlb multihit:         Not affected
L1tf:                  Not affected
Mds:                   Not affected
Meltdown:              Not affected
Mmio stale data:       Not affected
Reg file data sampling: Not affected
Retbleed:              Mitigation; Enhanced IBRS
Spec rstack overflow:  Not affected
Spec store bypass:     Mitigation; Speculative Store Bypass disabled via prctl
Spectre v1:            Mitigation; usercopy/swapgs barriers and __user pointer sanitization
Spectre v2:            Mitigation; Enhanced / Automatic IBRS; IBPB conditional; RSB filling; PBSRB-eIBRS SW sequence; BHI SW loop, KVM SW loop
Srbds:                 Not affected
Tsx async abort:       Not affected
```

```
user_rafi@DESKTOP-AC7MIM2:~$ lsblk
NAME
MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda   8:0    0 388.4M  1 disk
sdb   8:16   0   186M  1 disk
sdc   8:32   0    1G   0 disk [SWAP]
sdd   8:48   0    1T   0 disk /mnt/wslg/distro
/
```

1.1.2 Menggunakan Layanan OS

- Mempraktikkan pemanggilan layanan sistem (system calls) seperti proses, file, komunikasi, error handling, dan memori.
- Berfungsi untuk mendemonstrasikan berbagai aspek operasi sistem (OS services).

```
user_rafi@DESKTOP-AC7MIM2:~$ nano amiruddin.c
```

```
user_rafi@DESKTOP-AC7MIM2:~$ gcc -o amiruddin amiruddin.
user_rafi@DESKTOP-AC7MIM2:~$ ./amiruddin
=== Program Execution Service ===
Process ID (PID): 1956
Parent Process ID (PPID): 302

=== I/O Operations Service ===
File created and written successfully

=== File System Manipulation ===
File size: 23 bytes
File permissions: 644

=== Communications ===
Hostname: (null)

=== Error Detection ===
Error opening file: No such file or directory

=== Resource Allocation ===
Memory allocated successfully
```

1.2 User and OS Interface

1.2.1 Command Line Interface (CLI)

1.2.2 Program C - Shell Sederhana

- Berikut serangkaian aktivitas yang umum dilakukan dalam lingkungan Linux, termasuk navigasi sistem file, pembuatan direktori, manipulasi file, menjalankan proses di latar belakang, dan kompilasi serta eksekusi program C.
- menunjukkan berbagai perintah Linux yang digunakan untuk manipulasi file, proses, dan kompilasi kode. Ini adalah contoh tipikal dari bagaimana seorang pengguna berinteraksi dengan sistem operasi Linux melalui terminal.

```
user_rafi@DESKTOP-AC7MIM X + v
user_rafi@DESKTOP-AC7MIM2:~$ pwd
/home/user_rafi
user_rafi@DESKTOP-AC7MIM2:~$ ls -la
total 60
drwxr-x--- 4 user_rafi user_rafi 4096 Oct  8 19:06 .
drwxr-xr-x 3 root      root      4096 Oct  8 17:03 ..
-rw----- 1 user_rafi user_rafi   91 Oct  8 17:22 .bash_history
-rw-r--r-- 1 user_rafi user_rafi  220 Oct  8 17:03 .bash_logout
-rw-r--r-- 1 user_rafi user_rafi 3771 Oct  8 17:03 .bashrc
drwx----- 2 user_rafi user_rafi 4096 Oct  8 17:03 .cache
drwxr-xr-x 3 user_rafi user_rafi 4096 Oct  8 18:52 .local
-rw-rw-r-- 1 user_rafi user_rafi    0 Oct  8 17:03 .motd_shown
-rw-r--r-- 1 user_rafi user_rafi  807 Oct  8 17:03 .profile
-rw-r--r-- 1 user_rafi user_rafi    0 Oct  8 18:55 .sudo_as_admin_successful
-rwxr-xr-x 1 user_rafi user_rafi 16480 Oct  8 19:05 amiruddin
-rw-r--r-- 1 user_rafi user_rafi  1316 Oct  8 18:54 amiruddin.c
-rw-r--r-- 1 user_rafi user_rafi    23 Oct  8 19:05 test_file.txt
user_rafi@DESKTOP-AC7MIM2:~$ cd /tmp
user_rafi@DESKTOP-AC7MIM2:/tmp$ mkdir test_dir
user_rafi@DESKTOP-AC7MIM2:/tmp$ cd test_dir
user_rafi@DESKTOP-AC7MIM2:/tmp/test_dir$ cd
user_rafi@DESKTOP-AC7MIM2:~$ ls -l | grep "^d"
user_rafi@DESKTOP-AC7MIM2:~$ ps aux | grep bash | wc -l
3
user_rafi@DESKTOP-AC7MIM2:~$ echo "Hello World" > output.txt
user_rafi@DESKTOP-AC7MIM2:~$ cat output.txt
Hello World
user_rafi@DESKTOP-AC7MIM2:~$ cat output.txt >> output.txt
cat: output.txt: input file is output file
user_rafi@DESKTOP-AC7MIM2:~$ cat < output.txt
Hello World
user_rafi@DESKTOP-AC7MIM2:~$ sleep 30 &
[1] 1996
user_rafi@DESKTOP-AC7MIM2:~$ jobs
[1]+  Running                  sleep 30 &
user_rafi@DESKTOP-AC7MIM2:~$ fg %1
-bash: fg: job has terminated
[1]+  Done                    sleep 30
user_rafi@DESKTOP-AC7MIM2:~$ nano amiruddin2.c
user_rafi@DESKTOP-AC7MIM2:~$ gcc -o amiruddin2 amiruddin2.c
user_rafi@DESKTOP-AC7MIM2:~$ ^C
user_rafi@DESKTOP-AC7MIM2:~$ ./amiruddin2
myshell> help
```

1.3 System Calls

1.3.1 Process Control System Calls

1.3.2 File Management System Calls

1.3.3 Information Maintenance System Calls

- Program ini menampilkan layanan pemeliharaan informasi (Information Maintenance System Calls).
- Menunjukkan bagaimana sistem operasi menyimpan dan menyediakan informasi tentang proses, waktu, dan pengguna.

```
user_rafi@DESKTOP-AC7MIM X + v
user_rafi@DESKTOP-AC7MIM2:~$ nano amiruddin3.c
user_rafi@DESKTOP-AC7MIM2:~$ nano amiruddin4.c
user_rafi@DESKTOP-AC7MIM2:~$ nano amiruddin5.c
user_rafi@DESKTOP-AC7MIM2:~$ gcc -o amiruddin3 amiruddin3.c
user_rafi@DESKTOP-AC7MIM2:~$ gcc -o amiruddin4 amiruddin3,c
/usr/bin/ld: cannot find amiruddin3,c: No such file or directory
collect2: error: ld returned 1 exit status
user_rafi@DESKTOP-AC7MIM2:~$ gcc -o amiruddin4 amiruddin3.c
user_rafi@DESKTOP-AC7MIM2:~$ gcc -o amiruddin4 amiruddin4.c
user_rafi@DESKTOP-AC7MIM2:~$ gcc -o amiruddin5 amiruddin5.c
user_rafi@DESKTOP-AC7MIM2:~$ ./amiruddin3
=== Process Control System Calls ===
Parent PID: 2075
Parent waiting for child (PID: 2076)...
Child PID: 2076
Child's Parent PID: 2075
Child executing 'ls' command...
total 116
-rwxr-xr-x 1 user_rafi user_rafi 16480 Oct  8 19:05 amiruddin
-rw-r--r-- 1 user_rafi user_rafi 1316 Oct  8 18:54 amiruddin.c
-rwxr-xr-x 1 user_rafi user_rafi 16528 Oct  8 19:27 amiruddin2
-rw-r--r-- 1 user_rafi user_rafi  897 Oct  8 19:27 amiruddin2.c
-rwxr-xr-x 1 user_rafi user_rafi 16352 Oct  8 19:48 amiruddin3
-rw-r--r-- 1 user_rafi user_rafi  831 Oct  8 19:43 amiruddin3.c
-rwxr-xr-x 1 user_rafi user_rafi 16352 Oct  8 19:49 amiruddin4
-rw-r--r-- 1 user_rafi user_rafi  966 Oct  8 19:45 amiruddin4.c
-rwxr-xr-x 1 user_rafi user_rafi 16384 Oct  8 19:49 amiruddin5
-rw-r--r-- 1 user_rafi user_rafi  711 Oct  8 19:47 amiruddin5.c
-rw-r--r-- 1 user_rafi user_rafi  12 Oct  8 19:23 output.txt
-rw-r--r-- 1 user_rafi user_rafi   23 Oct  8 19:05 test_file.txt
Child exited with status: 0
user_rafi@DESKTOP-AC7MIM2:~$ ./amiruddin4
=== File Management System Calls ===
File descriptor: 3
Bytes written: 25
File closed
Read from file: Hello from system calls!
```

```
user_rafi@DESKTOP-AC7MIM X + v
user_rafi@DESKTOP-AC7MIM2:~$ ./amiruddin5
=== Information Maintenance System Calls ===

Process ID: 2078
Parent Process ID: 302
User ID: 1000
Group ID: 1000
Current time: Wed Oct 8 19:51:27 2025
Clock ticks since boot: 1718308175
Sleeping for 2 seconds...
Awake!
user_rafi@DESKTOP-AC7MIM2:~$ systemctl list-units --type=service
UNIT                                LOAD    ACTIVE SUB    DESCRIPTION
console-getty.service              loaded active running Console Getty
console-setup.service              loaded active exited Set console font and keymap
cron.service                       loaded active running Regular background program processing daemon
dbus.service                      loaded active running D-Bus System Message Bus
getty@tty1.service                 loaded active running Getty on tty1
keyboard-setup.service             loaded active exited Set the console keyboard layout
kmod-static-nodes.service          loaded active exited Create List of Static Device Nodes
polkit.service                    loaded active running Authorization Manager
rsyslog.service                   loaded active running System Logging Service
setvtrgb.service                  loaded active exited Set console scheme
snapd.seeded.service              loaded active exited Wait until snapd is fully seeded
systemd-binfmt.service            loaded active exited Set Up Additional Binary Formats
systemd-journal-flush.service      loaded active exited Flush Journal to Persistent Storage
systemd-journald.service           loaded active running Journal Service
systemd-logind.service            loaded active running User Login Management
systemd-modules-load.service       loaded active exited Load Kernel Modules
systemd-remount-fs.service         loaded active exited Remount Root and Kernel File Systems
systemd-resolved.service           loaded active running Network Name Resolution
systemd-sysctl.service            loaded active exited Apply Kernel Variables
systemd-timesyncd.service          loaded active running Network Time Synchronization
systemd-tmpfiles-setup-dev-early.service loaded active exited Create Static Device Nodes in /dev gracefully
systemd-tmpfiles-setup-dev.service loaded active exited Create Static Device Nodes in /dev
systemd-tmpfiles-setup.service     loaded active exited Create Volatile Files and Directories
systemd-udev-trigger.service       loaded active exited Coldplug All udev Devices
systemd-udevd.service              loaded active running Rule-based Manager for Device Events and Files
systemd-update-utmp.service        loaded active exited Record System Boot/Shutdown in UTMP
systemd-user-sessions.service      loaded active exited Permit User Sessions
unattended-upgrades.service        loaded active running Unattended Upgrades Shutdown
user-runtime-dir@1000.service      loaded active exited User Runtime Directory /run/user/1000
user@1000.service                 loaded active running User Manager for UID 1000
wsl-pro.service                   loaded active running Bridge to Ubuntu Pro agent on Windows

Legend: LOAD → Reflects whether the unit definition was properly loaded.
lines 1-34...skipping...
```


1.4 System Services

1.4.1 Menggunakan system

1.4.2 Membuat Custom System Service

- Gambar Berikut menunjukkan bagaimana menggunakan perintah systemctl untuk melihat daftar layanan yang berjalan di sistem Linux. Ini adalah cara yang berguna untuk memahami layanan apa yang diaktifkan dan berjalan, serta untuk memantau status sistem.

```
user_rafi@DESKTOP-AC7MIM2:~$ systemctl list-units --type=service
```

UNIT	LOAD	ACTIVE	SUB	DESCRIPTION
console-getty.service	loaded	active	running	Console Getty
console-setup.service	loaded	active	exited	Set console font and keymap
cron.service	loaded	active	running	Regular background program processing daemon
dbus.service	loaded	active	running	D-Bus System Message Bus
getty@tty1.service	loaded	active	running	Getty on tty1
keyboard-setup.service	loaded	active	exited	Set the console keyboard layout
kmod-static-nodes.service	loaded	active	exited	Create List of Static Device Nodes
polkit.service	loaded	active	running	Authorization Manager
rsyslog.service	loaded	active	running	System Logging Service
setvtrgb.service	loaded	active	exited	Set console scheme
snapd.seeded.service	loaded	active	exited	Wait until snapd is fully seeded
systemd-binfmt.service	loaded	active	exited	Set Up Additional Binary Formats
systemd-journal-flush.service	loaded	active	exited	Flush Journal to Persistent Storage

- Gambar Berikut Menunjukkan menunjukkan bagaimana menggunakan perintah systemctl dan journalctl untuk memantau status dan log dari layanan dbus. Ini adalah cara yang berguna untuk memecahkan masalah atau memahami bagaimana layanan beroperasi.

```
user_rafi@DESKTOP-AC7MIM2:~$ systemctl status dbus
```

● dbus.service - D-Bus System Message Bus

Loaded: loaded (/usr/lib/systemd/system/dbus.service; static)

Active: **active (running)** since Wed 2025-10-08 18:47:42 WIB; 3h 24min ago

TriggeredBy: ● dbus.socket

Docs: man:dbus-daemon(1)

Main PID: 169 (dbus-daemon)

Tasks: 1 (limit: 4532)

Memory: 2.3M (peak: 3.2M)

CPU: 1.157s

CGroup: /system.slice/dbus.service

└─169 @dbus-daemon --system --address=systemd: --nofork --nopidfile --systemd-ac

```
Oct 08 18:47:42 DESKTOP-AC7MIM2 systemd[1]: Starting dbus.service - D-Bus System Message Bus.
Oct 08 18:47:42 DESKTOP-AC7MIM2 systemd[1]: Started dbus.service - D-Bus System Message Bus.
Oct 08 18:47:43 DESKTOP-AC7MIM2 dbus-daemon[169]: [system] Activating via systemd: service na
Oct 08 18:47:43 DESKTOP-AC7MIM2 dbus-daemon[169]: [system] Successfully activated service 'or
Oct 08 18:57:41 DESKTOP-AC7MIM2 dbus-daemon[169]: [system] Activating via systemd: service na
Oct 08 18:57:41 DESKTOP-AC7MIM2 dbus-daemon[169]: [system] Activating via systemd: service na
Oct 08 18:57:41 DESKTOP-AC7MIM2 dbus-daemon[169]: [system] Successfully activated service 'or
Oct 08 18:57:41 DESKTOP-AC7MIM2 dbus-daemon[169]: [system] Successfully activated service 'or
Oct 08 21:29:36 DESKTOP-AC7MIM2 dbus-daemon[169]: [system] Activating via systemd: service na
Oct 08 21:29:36 DESKTOP-AC7MIM2 dbus-daemon[169]: [system] Successfully activated service 'or
[lines 1-22/22 (END)]
user_rafi@DESKTOP-AC7MIM2:~$ journalctl -u dbus -n 20
Oct 08 17:00:48 DESKTOP-AC7MIM2 systemd[1]: Starting dbus.service - D-Bus System Message Bus.
Oct 08 17:00:48 DESKTOP-AC7MIM2 systemd[1]: Started dbus.service - D-Bus System Message Bus.
Oct 08 17:00:50 DESKTOP-AC7MIM2 dbus-daemon[198]: [system] Activating via systemd: service na
Oct 08 17:00:50 DESKTOP-AC7MIM2 dbus-daemon[198]: [system] Successfully activated service 'or
```

- Gambar menunjukkan bagaimana menggunakan perintah `systemctl` untuk melihat daftar layanan yang diaktifkan di sistem Linux. Ini adalah cara yang berguna untuk memahami layanan apa yang akan dimulai secara otomatis saat sistem boot, dan untuk mengelola layanan mana yang diaktifkan atau dinonaktifkan.

```
user_rafi@DESKTOP-AC7MIM2:~$ systemctl list-unit-files --type=service --state=enabled
```

UNIT FILE	STATE	PRESET
apparmor.service	enabled	enabled
apport.service	enabled	enabled
cloud-config.service	enabled	enabled
cloud-final.service	enabled	enabled
cloud-init-local.service	enabled	enabled
cloud-init.service	enabled	enabled
console-setup.service	enabled	enabled
cron.service	enabled	enabled
dmesg.service	enabled	enabled
e2scrub_reap.service	enabled	enabled
getty@.service	enabled	enabled
keyboard-setup.service	enabled	enabled
landscape-client.service	enabled	enabled
networkd-dispatcher.service	enabled	enabled
rsyslog.service	enabled	enabled
setvtrgb.service	enabled	enabled
sshd.service	enabled	enabled

- Gambar menunjukkan upaya untuk membuat dan menjalankan layanan `systemd` kustom. Layanan tersebut gagal dimulai karena kesalahan dalam konfigurasi atau kode layanan. Kode kesalahan 203 menunjukkan bahwa ada masalah dengan file yang dapat dieksekusi yang ditentukan dalam file unit layanan. Log menunjukkan bahwa `systemd` mencoba untuk memulai ulang layanan, tetapi upaya tersebut terjadi terlalu cepat. Langkah selanjutnya adalah memeriksa file unit layanan dan kode daemon untuk mengidentifikasi dan memperbaiki kesalahan. Kemungkinan penyebab termasuk jalur yang salah ke file yang dapat dieksekusi, izin yang salah, atau kesalahan dalam kode daemon itu sendiri.

```

user_rafi@DESKTOP-AC7MIM2:~$ nano my_daemon.c
user_rafi@DESKTOP-AC7MIM2:~$ gcc -o my_daemon my_daemon.c
user_rafi@DESKTOP-AC7MIM2:~$ sudo nano /etc/systemd/system/my_daemon.service
[sudo] password for user_rafi:
user_rafi@DESKTOP-AC7MIM2:~$ sudo systemctl daemon-reload
user_rafi@DESKTOP-AC7MIM2:~$ sudo systemctl start my_daemon
Job for my_daemon.service failed because the control process exited with error code.
See "systemctl status my_daemon.service" and "journalctl -xeu my_daemon.service" for details.
user_rafi@DESKTOP-AC7MIM2:~$ sudo systemctl status my_daemon
* my_daemon.service - My Custom Daemon
   Loaded: loaded (/etc/systemd/system/my_daemon.service; disabled; preset: enabled)
   Active: failed (Result: exit-code) since Wed 2025-10-08 22:14:45 WIB; 7s ago
   Process: 2960 ExecStart=/path/to/my_daemon (code=exited, status=203/EXEC)
   CPU: 904us

Oct 08 22:14:45 DESKTOP-AC7MIM2 systemd[1]: my_daemon.service: Scheduled restart job, restart counter is at 5.
Oct 08 22:14:45 DESKTOP-AC7MIM2 systemd[1]: my_daemon.service: Start request repeated too quickly.
Oct 08 22:14:45 DESKTOP-AC7MIM2 systemd[1]: my_daemon.service: Failed with result 'exit-code'.
Oct 08 22:14:45 DESKTOP-AC7MIM2 systemd[1]: Failed to start my_daemon.service - My Custom Daemon.
user_rafi@DESKTOP-AC7MIM2:~$ sudo journalctl -u my_daemon
Oct 08 22:14:44 DESKTOP-AC7MIM2 systemd[1]: Starting my_daemon.service - My Custom Daemon...
Oct 08 22:14:44 DESKTOP-AC7MIM2 systemd[1]: my_daemon.service: Control process exited, code=exited, status=203/EXEC
Oct 08 22:14:44 DESKTOP-AC7MIM2 systemd[1]: my_daemon.service: Failed with result 'exit-code'.
Oct 08 22:14:44 DESKTOP-AC7MIM2 systemd[1]: Failed to start my_daemon.service - My Custom Daemon.
Oct 08 22:14:44 DESKTOP-AC7MIM2 systemd[1]: my_daemon.service: Scheduled restart job, restart counter is at 1.
Oct 08 22:14:44 DESKTOP-AC7MIM2 systemd[1]: Starting my_daemon.service - My Custom Daemon...

```

- Gambar menunjukkan upaya untuk menghentikan layanan systemd yang sebelumnya gagal dijalankan. Ini adalah langkah yang logis setelah layanan gagal dimulai, untuk mencegah upaya restart yang tidak perlu. Setelah layanan dihentikan, langkah selanjutnya adalah memperbaiki konfigurasi atau kode layanan, dan kemudian mencoba memulainya kembali.

```

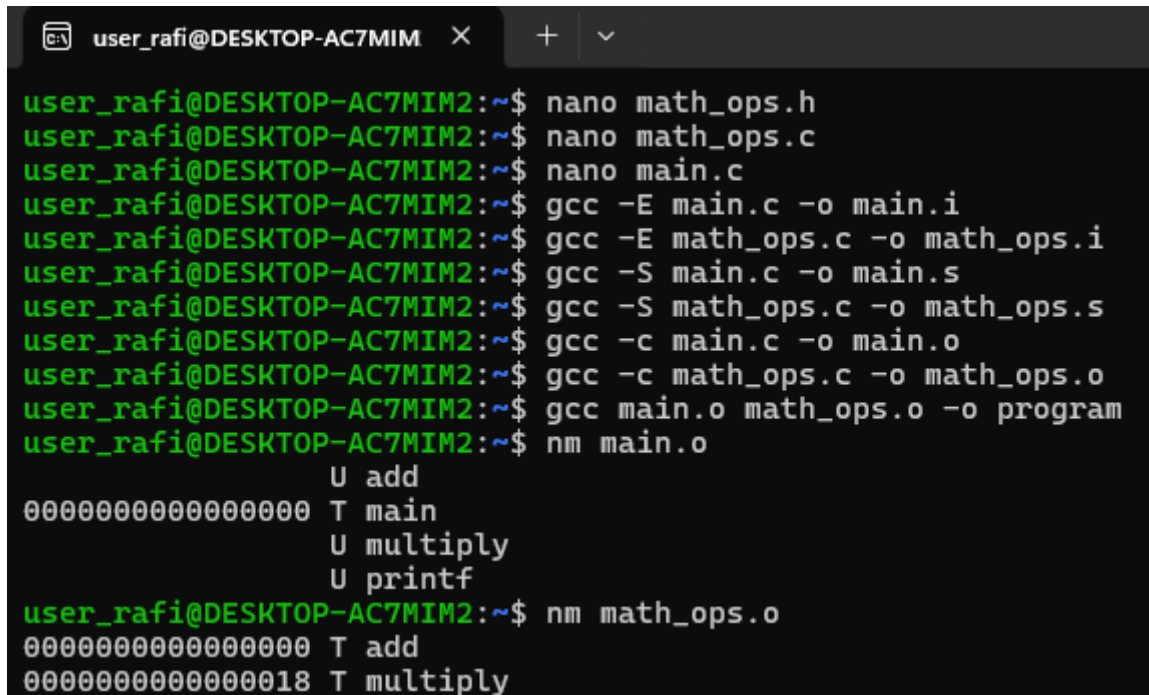
Oct 08 22:14:45 DESKTOP-AC7MIM2 systemd[1]: Starting my_daemon.service - My Custom Daemon...
Oct 08 22:14:45 DESKTOP-AC7MIM2 systemd[1]: my_daemon.service: Control process exited, code=exited, status=203/EXEC
Oct 08 22:14:45 DESKTOP-AC7MIM2 systemd[1]: my_daemon.service: Failed with result 'exit-code'.
Oct 08 22:14:45 DESKTOP-AC7MIM2 systemd[1]: Failed to start my_daemon.service - My Custom Daemon.
Oct 08 22:14:45 DESKTOP-AC7MIM2 systemd[1]: my_daemon.service: Scheduled restart job, restart counter is at 5.
Oct 08 22:14:45 DESKTOP-AC7MIM2 systemd[1]: my_daemon.service: Start request repeated too quickly.
Oct 08 22:14:45 DESKTOP-AC7MIM2 systemd[1]: my_daemon.service: Failed with result 'exit-code'.
Oct 08 22:14:45 DESKTOP-AC7MIM2 systemd[1]: Failed to start my_daemon.service - My Custom Daemon.
user_rafi@DESKTOP-AC7MIM2:~$ sudo systemctl stop my_daemon

```

1.5 Linkers and Loaders

1.5.1 Proses Linking

- Gambar menunjukkan proses kompilasi program C yang modular, yang melibatkan pemisahan kode ke dalam beberapa file sumber dan penggunaan Header File. Alat nm digunakan untuk memeriksa simbol-simbol dalam file objek, yang menunjukkan fungsi mana yang didefinisikan dan digunakan dalam setiap file. Proses kompilasi melibatkan beberapa tahap, termasuk Preprocessing, kompilasi ke Assembly, kompilasi ke file objek, dan linking untuk membuat program yang dapat dieksekusi.



```
user_rafi@DESKTOP-AC7MIM2:~$ nano math_ops.h
user_rafi@DESKTOP-AC7MIM2:~$ nano math_ops.c
user_rafi@DESKTOP-AC7MIM2:~$ nano main.c
user_rafi@DESKTOP-AC7MIM2:~$ gcc -E main.c -o main.i
user_rafi@DESKTOP-AC7MIM2:~$ gcc -E math_ops.c -o math_ops.i
user_rafi@DESKTOP-AC7MIM2:~$ gcc -S main.c -o main.s
user_rafi@DESKTOP-AC7MIM2:~$ gcc -S math_ops.c -o math_ops.s
user_rafi@DESKTOP-AC7MIM2:~$ gcc -c main.c -o main.o
user_rafi@DESKTOP-AC7MIM2:~$ gcc -c math_ops.c -o math_ops.o
user_rafi@DESKTOP-AC7MIM2:~$ gcc main.o math_ops.o -o program
user_rafi@DESKTOP-AC7MIM2:~$ nm main.o
                 U add
0000000000000000 T main
                 U multiply
                 U printf
user_rafi@DESKTOP-AC7MIM2:~$ nm math_ops.o
0000000000000000 T add
0000000000000018 T multiply
```

- Gambar menunjukkan analisis program yang dapat dieksekusi setelah proses kompilasi dan linking. Alat nm digunakan untuk memahami struktur internal program dan simbol-simbol yang digunakan, dan alat ldd digunakan untuk memahami dependensi program pada shared libraries. Informasi ini penting untuk debugging, optimasi, dan deployment program.

```

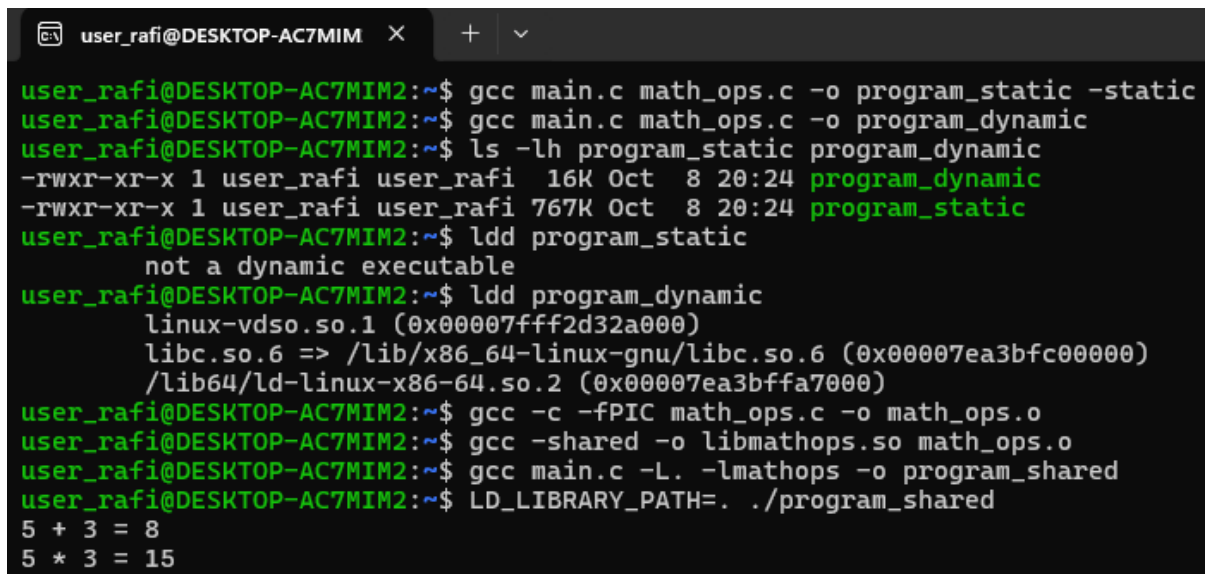
user_rafi@DESKTOP-AC7MIM2:~$ nm program
0000000000003dc8 d _DYNAMIC
0000000000003fb8 d _GLOBAL_OFFSET_TABLE_
0000000000002000 R _IO_stdin_used
                 w _ITM_deregisterTMCloneTable
                 w _ITM_registerTMCloneTable
0000000000002148 r __FRAME_END__
000000000000201c r __GNU_EH_FRAME_HDR
0000000000004010 D __TMC_END__
00000000000038c r __abi_tag
0000000000004010 B __bss_start
                 w __cxa_finalize@GLIBC_2.2.5
0000000000004000 D __data_start
0000000000001100 t __do_global_dtors_aux
0000000000003dc0 d __do_global_dtors_aux_fini_array_entry
0000000000004008 D __dso_handle
0000000000003db8 d __frame_dummy_init_array_entry
                 w __gmon_start__
                 U __libc_start_main@GLIBC_2.34
0000000000004010 D _edata
0000000000004018 B _end
00000000000011d4 T _fini
0000000000001000 T _init
0000000000001060 T _start
00000000000011a2 T add
0000000000004010 b completed.0
0000000000004000 W data_start
0000000000001090 t deregister_tm_clones
0000000000001140 t frame_dummy
0000000000001149 T main
00000000000011ba T multiply
                 U printf@GLIBC_2.2.5
00000000000010c0 t register_tm_clones
user_rafi@DESKTOP-AC7MIM2:~$ ldd program
        linux-vdso.so.1 (0x00007fff60d33000)
        libc.so.6 => /lib/x86_64-linux-gnu/libc.so.6 (0x00007d08c8a00000)
        /lib64/ld-linux-x86-64.so.2 (0x00007d08c8c31000)

```

1.5.2 Static vs Dynamic Linking

1.5.3 Membuat dan Menggunakan Shared Library

- Gambar menunjukkan perbedaan antara static linking dan dynamic linking dalam kompilasi program C. static linking menghasilkan executable yang lebih besar tetapi tidak bergantung pada shared libraries, sementara dynamic linking menghasilkan executable yang lebih kecil tetapi bergantung pada shared libraries. Gambar juga menunjukkan cara membuat dan menggunakan shared library kustom.



```
user_rafi@DESKTOP-AC7MIM:~$ gcc main.c math_ops.c -o program_static -static
user_rafi@DESKTOP-AC7MIM:~$ gcc main.c math_ops.c -o program_dynamic
user_rafi@DESKTOP-AC7MIM:~$ ls -lh program_static program_dynamic
-rwxr-xr-x 1 user_rafi user_rafi 16K Oct  8 20:24 program_dynamic
-rwxr-xr-x 1 user_rafi user_rafi 767K Oct  8 20:24 program_static
user_rafi@DESKTOP-AC7MIM:~$ ldd program_static
not a dynamic executable
user_rafi@DESKTOP-AC7MIM:~$ ldd program_dynamic
linux-vdso.so.1 (0x00007fff2d32a000)
libc.so.6 => /lib/x86_64-linux-gnu/libc.so.6 (0x00007ea3bfc00000)
/lib64/ld-linux-x86-64.so.2 (0x00007ea3bffa7000)
user_rafi@DESKTOP-AC7MIM:~$ gcc -c -fPIC math_ops.c -o math_ops.o
user_rafi@DESKTOP-AC7MIM:~$ gcc -shared -o libmathops.so math_ops.o
user_rafi@DESKTOP-AC7MIM:~$ gcc main.c -L. -lmathops -o program_shared
user_rafi@DESKTOP-AC7MIM:~$ LD_LIBRARY_PATH=. ./program_shared
5 + 3 = 8
5 * 3 = 15
```

- Gambar menunjukkan analisis bagian dynamic dari program yang di link secara dynamic, menggunakan alat readelf. Informasi ini memberikan wawasan tentang bagaimana program berinteraksi dengan dynamic linker dan shared libraries saat dijalankan. Ini termasuk libraries mana yang dibutuhkan, fungsi inisialisasi dan finalisasi, dan flagdynamic linker.

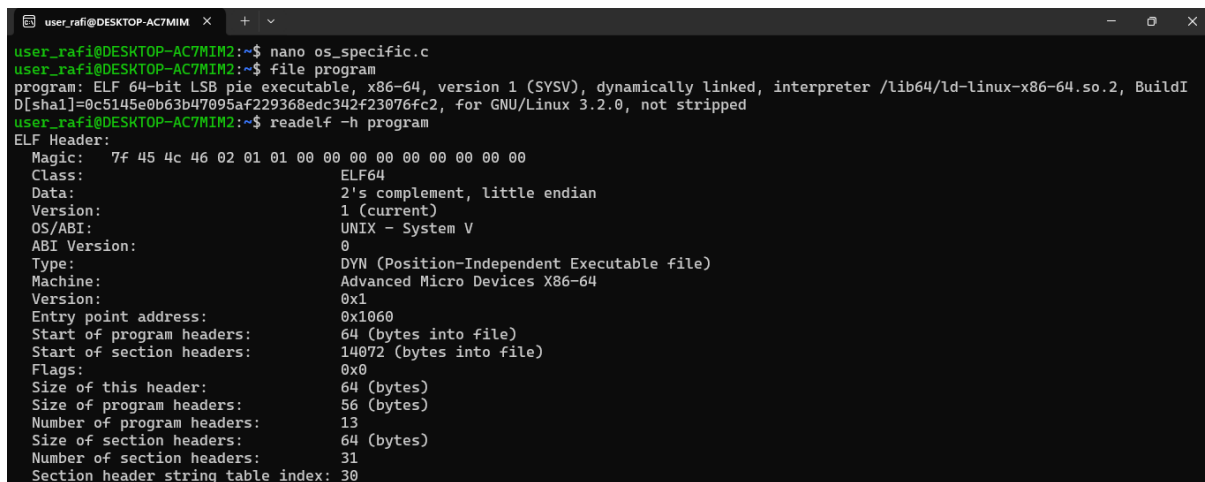
```
user_rafi@DESKTOP-AC7MIM2:~$ readelf -d program_shared

Dynamic section at offset 0x2da8 contains 28 entries:
   Tag               Type                             Name/Value
0x0000000000000001 (NEEDED)           Shared library: [libmathops.so]
0x0000000000000001 (NEEDED)           Shared library: [libc.so.6]
0x000000000000000c (INIT)             0x1000
0x000000000000000d (FINI)             0x11e4
0x0000000000000019 (INIT_ARRAY)        0x3d98
0x000000000000001b (INIT_ARRAYSZ)      8 (bytes)
0x000000000000001a (FINI_ARRAY)        0x3da0
0x000000000000001c (FINI_ARRAYSZ)      8 (bytes)
0x0000000006ffffef5 (GNU_HASH)         0x3b0
0x0000000000000005 (STRTAB)            0x4b0
0x0000000000000006 (SYMTAB)            0x3d8
0x000000000000000a (STRSZ)             170 (bytes)
0x000000000000000b (SYMENT)            24 (bytes)
0x0000000000000015 (DEBUG)             0x0
0x0000000000000003 (PLTGOT)            0x3fa8
0x0000000000000002 (PLTRELSZ)          72 (bytes)
0x0000000000000014 (PLTREL)            RELA
0x0000000000000017 (JMPREL)            0x660
0x0000000000000007 (RELA)              0x5a0
0x0000000000000008 (RELASZ)            192 (bytes)
0x0000000000000009 (RELAENT)           24 (bytes)
0x000000000000001e (FLAGS)             BIND_NOW
0x0000000006fffffb (FLAGS_1)           Flags: NOW PIE
0x0000000006fffffe (VERNEED)           0x570
0x0000000006ffffff (VERNEEDNUM)        1
0x0000000006ffffff0 (VERSYM)           0x55a
0x0000000006ffffff9 (RELACOUNT)        3
0x0000000000000000 (NULL)              0x0
user_rafi@DESKTOP-AC7MIM2:~$ |
```

1.6 OS-Specific Applications

1.6.1 System Call Differences

- Gambar menunjukkan analisis executable menggunakan alat file dan readelf untuk memahami format dan karakteristik executable tersebut. Alat file memberikan ringkasan singkat tentang jenis executable, sementara alat readelf memberikan informasi detail tentang header ELF, yang memberikan wawasan tentang struktur dan konfigurasi executable.



```
user_rafi@DESKTOP-AC7MIM:~$ nano os_specific.c
user_rafi@DESKTOP-AC7MIM:~$ file program
program: ELF 64-bit LSB pie executable, x86-64, version 1 (SYSV), dynamically linked, interpreter /lib64/ld-linux-x86-64.so.2, BuildID[sha1]=0c5145e0b63b47095af229368edc342f23076fc2, for GNU/Linux 3.2.0, not stripped
user_rafi@DESKTOP-AC7MIM:~$ readelf -h program
ELF Header:
  Magic:   7f 45 4c 46 02 01 01 00 00 00 00 00 00 00 00 00
  Class:             ELF64
  Data:              2's complement, little endian
  Version:           1 (current)
  OS/ABI:            UNIX - System V
  ABI Version:       0
  Type:              DYN (Position-Independent Executable file)
  Machine:           Advanced Micro Devices X86-64
  Version:           0x1
  Entry point address: 0x1060
  Start of program headers: 64 (bytes into file)
  Start of section headers: 14072 (bytes into file)
  Flags:             0x0
  Size of this header: 64 (bytes)
  Size of program headers: 56 (bytes)
  Number of program headers: 13
  Size of section headers: 64 (bytes)
  Number of section headers: 31
  Section header string table index: 30
```

1.6.2 Binary Format Analysis

- Gambar menunjukkan analisis section header dari executable menggunakan alat readelf. Informasi ini memberikan wawasan tentang bagaimana executable diorganisasikan ke dalam berbagai bagian, yang masing-masing memiliki tujuan dan karakteristik sendiri. Ini membantu dalam memahami struktur internal executable dan bagaimana executable berinteraksi dengan sistem operasi dan dynamic linker.

```
user_rafi@DESKTOP-AC7MIM2:~$ readelf -S program
There are 31 section headers, starting at offset 0x36f8:

Section Headers:
```

[Nr]	Name	Type	Address	Offset
	Size	EntSize	Flags Link Info	Align
[0]	0000000000000000	NULL	0000000000000000	00000000
[1]	.interp	PROGBITS	0000000000000318	00000318
[2]	.note.gnu.pr[...]	NOTE	0000000000000338	00000338
[3]	.note.gnu.bu[...]	NOTE	0000000000000368	00000368
[4]	.note.ABI-tag	NOTE	000000000000038c	0000038c
[5]	.gnu.hash	GNU_HASH	00000000000003b0	000003b0
[6]	.dynsym	DYNSYM	00000000000003d8	000003d8
[7]	.dynstr	STRTAB	0000000000000480	00000480
[8]	.gnu.version	VERSYM	0000000000000510	00000510
[9]	.gnu.version_r	VERNEED	0000000000000520	00000520
[10]	.rela.dyn	RELA	0000000000000550	00000550
[11]	.rela.plt	RELA	0000000000000610	00000610
[12]	.init	PROGBITS	0000000000001000	00001000

- Gambar menunjukkan analisis program headers dari executable menggunakan alat readelf. Informasi ini memberikan wawasan tentang bagaimana sistem operasi load dan menjalankan executable, termasuk segmen memori mana yang perlu dibuat, di mana kode dan data berada, dan izin apa yang harus diberikan. Program headers sangat penting untuk memahami bagaimana executable berinteraksi dengan sistem operasi.

```
user_rafi@DESKTOP-AC7MIM2:~$ readelf -l program

Elf file type is DYN (Position-Independent Executable file)
Entry point 0x1060
There are 13 program headers, starting at offset 64

Program Headers:
  Type           Offset             VirtAddr           PhysAddr
                 FileSiz             MemSiz              Flags   Align
PHDR              0x0000000000000040 0x0000000000000040 0x0000000000000040
                 0x00000000000002d8 0x00000000000002d8  R       0x8
INTERP            0x0000000000000318 0x0000000000000318 0x0000000000000318
                 0x000000000000001c 0x000000000000001c  R       0x1
  [Requesting program interpreter: /lib64/ld-linux-x86-64.so.2]
LOAD              0x0000000000000000 0x0000000000000000 0x0000000000000000
                 0x0000000000000628 0x0000000000000628  R       0x1000
LOAD              0x0000000000000100 0x0000000000000100 0x0000000000000100
                 0x00000000000001e1 0x00000000000001e1  R E     0x1000
LOAD              0x0000000000000200 0x0000000000000200 0x0000000000000200
                 0x000000000000014c 0x000000000000014c  R       0x1000
LOAD              0x00000000000002db8 0x00000000000003db8 0x00000000000003db8
                 0x0000000000000258 0x0000000000000260  RW      0x1000
DYNAMIC            0x00000000000002dc8 0x00000000000003dc8 0x00000000000003dc8
                 0x00000000000001f0 0x00000000000001f0  RW      0x8
NOTE              0x0000000000000338 0x0000000000000338 0x0000000000000338
                 0x0000000000000030 0x0000000000000030  R       0x8
NOTE              0x0000000000000368 0x0000000000000368 0x0000000000000368
                 0x0000000000000044 0x0000000000000044  R       0x4
GNU_PROPERTY       0x0000000000000338 0x0000000000000338 0x0000000000000338
                 0x0000000000000030 0x0000000000000030  R       0x8
GNU_EH_FRAME      0x0000000000000201c 0x0000000000000201c 0x0000000000000201c
                 0x0000000000000044 0x0000000000000044  R       0x4
GNU_STACK         0x0000000000000000 0x0000000000000000 0x0000000000000000
                 0x0000000000000000 0x0000000000000000  RW      0x10
```

- Gambar menunjukkan disassembly dari executable menggunakan alat objdump. Informasi ini memberikan wawasan tentang instruksi-instruksi assembly yang membentuk program, yang berguna untuk debugging, analisis keamanan, dan pemahaman tentang bagaimana program beroperasi pada tingkat rendah.

```

user_rafi@DESKTOP-AC7MIM2:~$ objdump -d program | head -50

program:      file format elf64-x86-64

Disassembly of section .init:

0000000000001000 <_init>:
   1000:    f3 0f 1e fa                endbr64
   1004:    48 83 ec 08                sub     $0x8,%rsp
   1008:    48 8b 05 d9 2f 00 00      mov     0x2fd9(%rip),%rax        # 3fe8 <__gmon_start__@Base>
   100f:    48 85 c0                  test    %rax,%rax
   1012:    74 02                      je      1016 <_init+0x16>
   1014:    ff d0                      call    *%rax
   1016:    48 83 c4 08                add     $0x8,%rsp
   101a:    c3                        ret

Disassembly of section .plt:

0000000000001020 <.plt>:
   1020:    ff 35 9a 2f 00 00      push    0x2f9a(%rip)            # 3fc0 <_GLOBAL_OFFSET_TABLE_+0x8>
   1026:    ff 25 9c 2f 00 00      jmp     *0x2f9c(%rip)          # 3fc8 <_GLOBAL_OFFSET_TABLE_+0x10>
   102c:    0f 1f 40 00            nopl    0x0(%rax)
   1030:    f3 0f 1e fa                endbr64
   1034:    68 00 00 00 00          push    $0x0
   1039:    e9 e2 ff ff            jmp     1020 <_init+0x20>
   103e:    66 90                    xchg    %ax,%ax

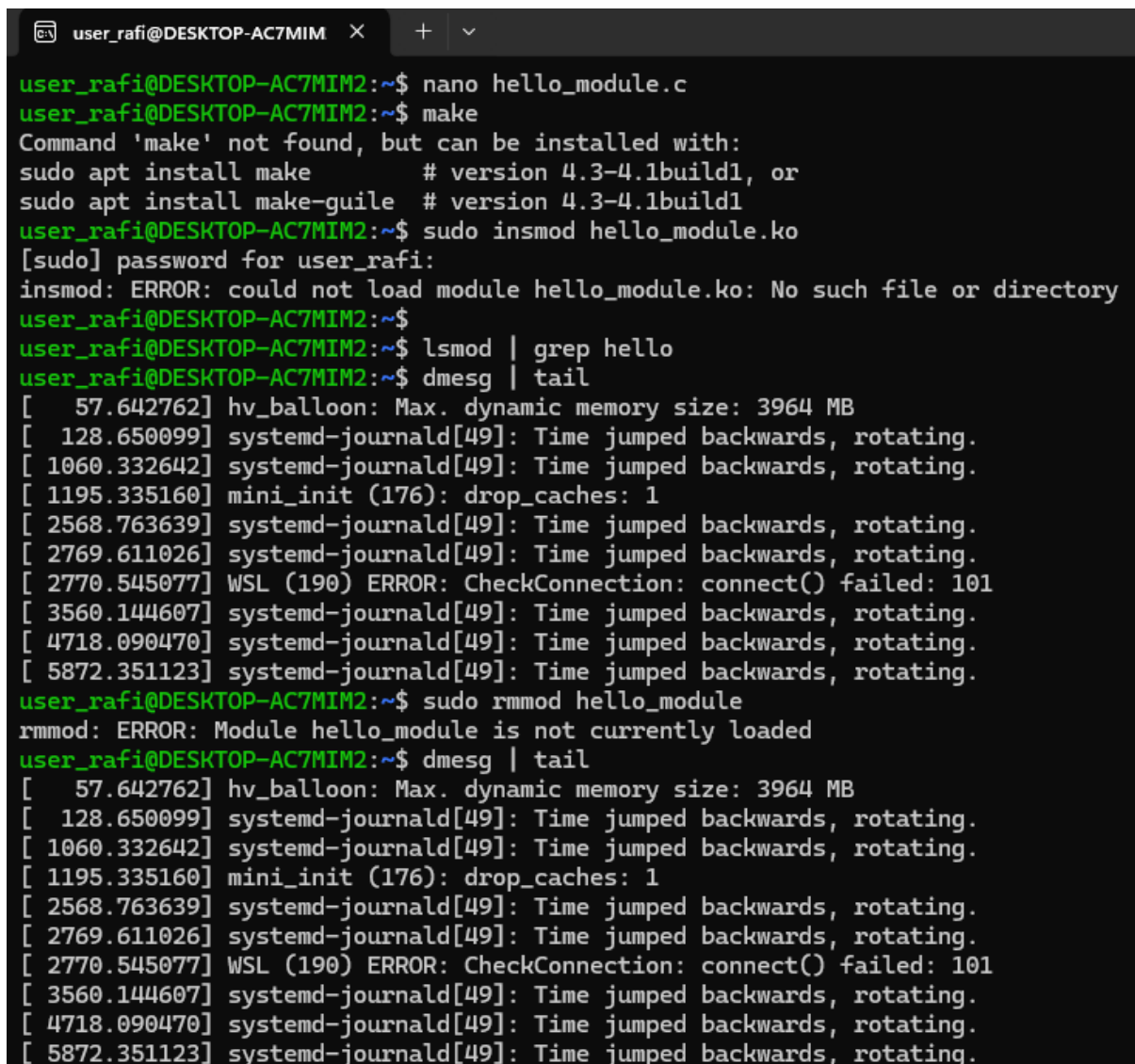
Disassembly of section .plt.got:

```

1.7 OS Design and Implementation

1.7.1 Kernel Module Sederhana

- Gambar menunjukkan upaya untuk membuat, load, dan unload kernel module. Awalnya, perintah make tidak ditemukan, tetapi ini diperbaiki dengan menginstalnya. Namun, kernel module gagal di load karena file hello_module.ko tidak ditemukan, yang menunjukkan bahwa kernel module belum berhasil dibuat. Karena kernel module tidak pernah berhasil di load, upaya untuk meng unload nya juga gagal. Langkah selanjutnya adalah memastikan bahwa kernel module berhasil dibuat menggunakan perintah make sebelum mencoba load nya. Perlu diperiksa makefile dan kode sumber hello_module.c untuk memastikan tidak ada kesalahan.



```
user_rafi@DESKTOP-AC7MIM2:~$ nano hello_module.c
user_rafi@DESKTOP-AC7MIM2:~$ make
Command 'make' not found, but can be installed with:
sudo apt install make          # version 4.3-4.1build1, or
sudo apt install make-guile    # version 4.3-4.1build1
user_rafi@DESKTOP-AC7MIM2:~$ sudo insmod hello_module.ko
[sudo] password for user_rafi:
insmod: ERROR: could not load module hello_module.ko: No such file or directory
user_rafi@DESKTOP-AC7MIM2:~$
user_rafi@DESKTOP-AC7MIM2:~$ lsmod | grep hello
user_rafi@DESKTOP-AC7MIM2:~$ dmesg | tail
[ 57.642762] hv_balloon: Max. dynamic memory size: 3964 MB
[ 128.650099] systemd-journald[49]: Time jumped backwards, rotating.
[ 1060.332642] systemd-journald[49]: Time jumped backwards, rotating.
[ 1195.335160] mini_init (176): drop_caches: 1
[ 2568.763639] systemd-journald[49]: Time jumped backwards, rotating.
[ 2769.611026] systemd-journald[49]: Time jumped backwards, rotating.
[ 2770.545077] WSL (190) ERROR: CheckConnection: connect() failed: 101
[ 3560.144607] systemd-journald[49]: Time jumped backwards, rotating.
[ 4718.090470] systemd-journald[49]: Time jumped backwards, rotating.
[ 5872.351123] systemd-journald[49]: Time jumped backwards, rotating.
user_rafi@DESKTOP-AC7MIM2:~$ sudo rmmod hello_module
rmmod: ERROR: Module hello_module is not currently loaded
user_rafi@DESKTOP-AC7MIM2:~$ dmesg | tail
[ 57.642762] hv_balloon: Max. dynamic memory size: 3964 MB
[ 128.650099] systemd-journald[49]: Time jumped backwards, rotating.
[ 1060.332642] systemd-journald[49]: Time jumped backwards, rotating.
[ 1195.335160] mini_init (176): drop_caches: 1
[ 2568.763639] systemd-journald[49]: Time jumped backwards, rotating.
[ 2769.611026] systemd-journald[49]: Time jumped backwards, rotating.
[ 2770.545077] WSL (190) ERROR: CheckConnection: connect() failed: 101
[ 3560.144607] systemd-journald[49]: Time jumped backwards, rotating.
[ 4718.090470] systemd-journald[49]: Time jumped backwards, rotating.
[ 5872.351123] systemd-journald[49]: Time jumped backwards, rotating.
```

1.7.2 Monitoring Kernel

- Gambar menunjukkan penggunaan berbagai perintah untuk mendapatkan informasi tentang sistem operasi dan melihat log sistem. Perintah `uname` dan `cat /proc/version` memberikan informasi tentang versi kernel, sementara perintah `systctl` (yang gagal) menunjukkan upaya untuk melihat parameter kernel. Perintah `dmesg` digunakan untuk melihat pesan kernel, yang dapat membantu dalam debugging masalah sistem.

```
user_rafi@DESKTOP-AC7MIM X + v
user_rafi@DESKTOP-AC7MIM2:~$ uname -r
6.6.87.2-microsoft-standard-WSL2
user_rafi@DESKTOP-AC7MIM2:~$ cat /proc/version
Linux version 6.6.87.2-microsoft-standard-WSL2 (root@439a258ad544) (gcc (GCC) 11.2.0, GNU ld (GNU Binutils) 2.37) #1 SMP PREEMPT_DYNAMIC T
18:30:46 UTC 2025
user_rafi@DESKTOP-AC7MIM2:~$ systctl -a | head -20
systctl: permission denied on key 'kernel.cad_pid'
abi.vsyscall32 = 1
debug.exception-trace = 1
debug.kprobes-optimization = 1
dev.cdrom.autoclose = 1
dev.cdrom.autoeject = 0
dev.cdrom.check_media = 0
dev.cdrom.debug = 0
dev.cdrom.info = CD-ROM information, Id: cdrom.c 3.20 2003/12/17
dev.cdrom.info =
dev.cdrom.info = drive name:
dev.cdrom.info = drive speed:
dev.cdrom.info = drive # of slots:
dev.cdrom.info = Can close tray:
dev.cdrom.info = Can open tray:
dev.cdrom.info = Can lock tray:
dev.cdrom.info = Can change speed:
dev.cdrom.info = Can select disk:
dev.cdrom.info = Can read multisession:
dev.cdrom.info = Can read MCN:
dev.cdrom.info = Reports media changed:
user_rafi@DESKTOP-AC7MIM2:~$ dmesg | tail -30
[ 1.739392] EXT4-fs (sdd): mounted filesystem 2c11e05f-8896-4c3c-9e13-e3b5efc6162c r/w with ordered data mode. Quota mode: none.
[ 3.296202] pulseaudio[224]: memfd_create() called without MFD_EXEC or MFD_NOEXEC_SEAL set
[ 3.587909] Failed to connect to bus: No such file or directory
[ 3.943470] Failed to connect to bus: No such file or directory
[ 4.039997] misc dxg: dxgk: dxgkio_is_feature_enabled: Ioctl failed: -22
[ 4.134124] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -22
[ 4.135553] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -22
```

- Gambar menunjukkan penggunaan perintah `lsmod` untuk melihat daftar kernel module yang di load dan perintah `modinfo` untuk mendapatkan informasi tentang kernel module `ext4`. `ext4` adalah sistem file yang umum digunakan di Linux, dan informasi yang ditampilkan memberikan wawasan tentang bagaimana module tersebut terintegrasi ke dalam kernel.

```
[ 3560.144607] systemd-journald[49]: Time jumped backwards, rotating.
[ 4718.090470] systemd-journald[49]: Time jumped backwards, rotating.
[ 5872.351123] systemd-journald[49]: Time jumped backwards, rotating.
user_rafi@DESKTOP-AC7MIM2:~$ lsmod
Module                Size  Used by
tls                   98304  0
intel_rapl_msr        16384  0
intel_rapl_common     32768  1 intel_rapl_msr
kvm_intel             356352  0
kvm                   970752  1 kvm_intel
irqbypass             12288  1 kvm
crc32c_intel          16384  0
ac                     16384  0
battery               20480  0
sch_fq_codel          16384  1
configfs              53248  1
autofs4               45056  0
br_netfilter          28672  0
bridge                282624  1 br_netfilter
stp                   12288  1 bridge
llc                   12288  2 bridge,stp
ip_tables             28672  0
tun                   53248  0
user_rafi@DESKTOP-AC7MIM2:~$ modinfo ext4
name:                ext4
filename:             (builtin)
softdep:              pre: crc32c
license:              GPL
file:                 fs/ext4/ext4
description:          Fourth Extended Filesystem
author:               Remy Card, Stephen Tweedie, Andrew Morton, Andreas Dilger, Theodore Ts'o and others
alias:                fs-ext4
alias:                ext3
alias:                fs-ext3
```

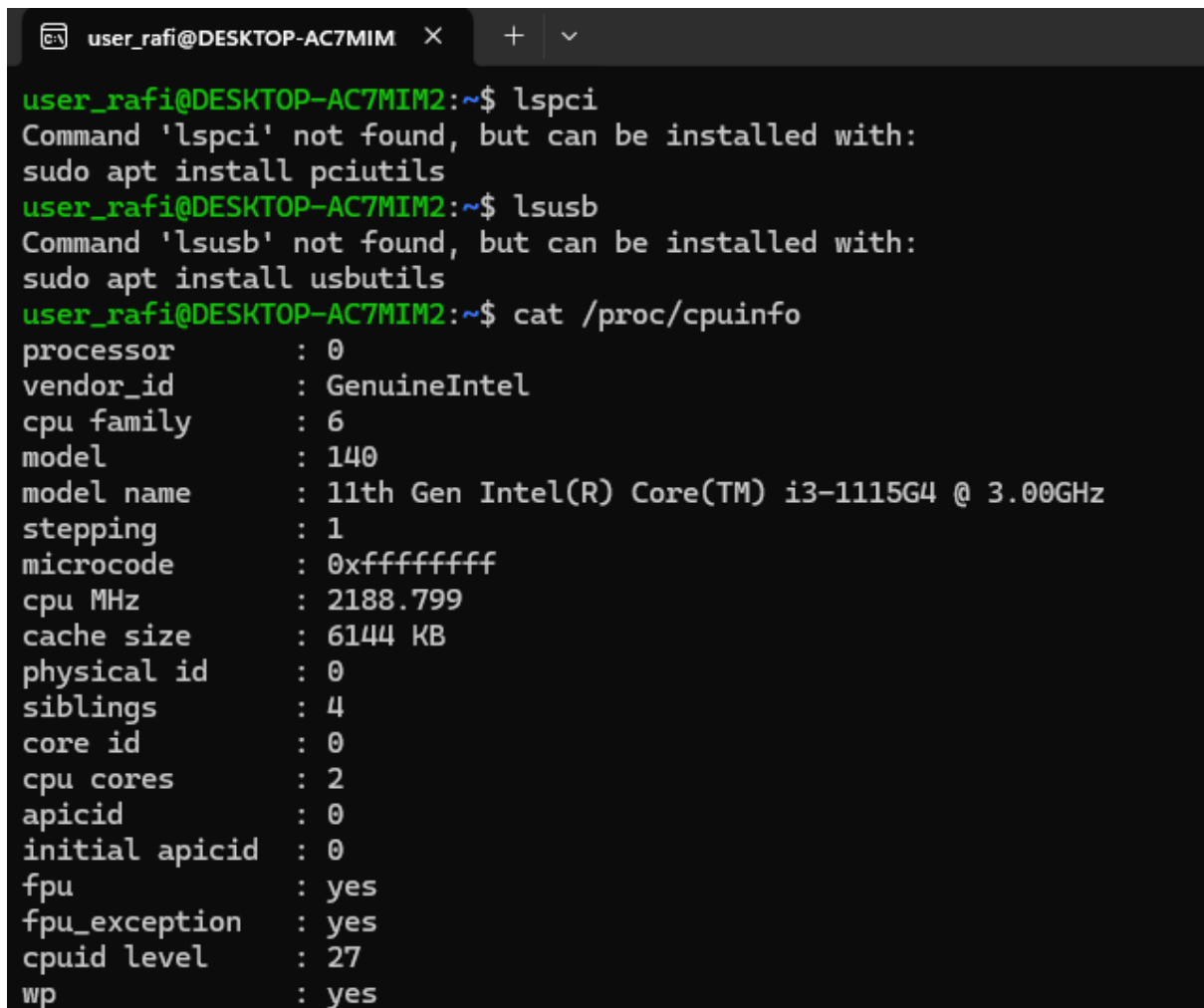
1.8 OS Structure

1.8.1 Eksplorasi Struktur Layering

1.8.2 Microkernel vs Monolithic

1.8.3 Program - Virtual File System

- Gambar dibawah ini menunjukkan upaya untuk mengaudit konfigurasi hardware sistem, namun hanya berhasil mendapatkan informasi detail prosesor karena utilitas lspci dan lsusb belum terinstal



```
user_rafi@DESKTOP-AC7MIM x + v
user_rafi@DESKTOP-AC7MIM2:~$ lspci
Command 'lspci' not found, but can be installed with:
sudo apt install pciutils
user_rafi@DESKTOP-AC7MIM2:~$ lsusb
Command 'lsusb' not found, but can be installed with:
sudo apt install usbutils
user_rafi@DESKTOP-AC7MIM2:~$ cat /proc/cpuinfo
processor          : 0
vendor_id         : GenuineIntel
cpu family        : 6
model             : 140
model name        : 11th Gen Intel(R) Core(TM) i3-1115G4 @ 3.00GHz
stepping          : 1
microcode         : 0xffffffff
cpu MHz           : 2188.799
cache size        : 6144 KB
physical id       : 0
siblings          : 4
core id           : 0
cpu cores         : 2
apicid            : 0
initial apicid    : 0
fpu               : yes
fpu_exception     : yes
cpuid level       : 27
wp               : yes
```

- Inti dari rangkaian perintah ini adalah untuk mendapatkan status sistem secara mendalam, mulai dari identitas *kernel* hingga daftar lengkap semua program dan layanan yang sedang aktif.

```
user_rafi@DESKTOP-AC7MIM2:~$ uname -a
Linux DESKTOP-AC7MIM2 6.6.87.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu Jun  5 18:30:46 UTC 2025 x86_64 x86_64 x86_64 GNU/Linux
user_rafi@DESKTOP-AC7MIM2:~$ cat /proc/sys/kernel/ostype
Linux
user_rafi@DESKTOP-AC7MIM2:~$ strace ls
Command 'strace' not found, but can be installed with:
sudo apt install strace
user_rafi@DESKTOP-AC7MIM2:~$ ps aux
```

USER	PID	%CPU	%MEM	VSZ	RSS	TTY	STAT	START	TIME	COMMAND
root	1	0.0	0.3	21688	12288	?	Ss	18:50	0:02	/usr/lib/systemd/systemd --system --deserialize=24
root	2	0.0	0.0	3072	1792	?	Sl	18:50	0:00	/init
root	7	0.0	0.0	3072	1792	?	Sl	18:50	0:00	plan9 --control-socket 7 --log-level 4 --server-fd 8 --pipe-fd 10 --log-truncate
root	49	0.0	0.4	66836	18884	?	S<s	18:50	0:01	/usr/lib/systemd/systemd-journald
root	102	0.0	0.1	25268	6272	?	Ss	18:50	0:00	/usr/lib/systemd/systemd-udevd
systemd+	117	0.0	0.3	21456	12416	?	Ss	18:50	0:00	/usr/lib/systemd/systemd-resolved
systemd+	118	0.0	0.1	91024	7552	?	Ssl	18:50	0:00	/usr/lib/systemd/systemd-timesyncd
root	168	0.0	0.0	4236	2560	?	Ss	18:50	0:00	/usr/sbin/cron -f -P
message+	169	0.0	0.1	9688	5248	?	Ss	18:50	0:00	@dbus-daemon --system --address=systemd: --nofork --nopidfile --systemd-activation
root	176	0.0	0.2	17960	8576	?	Ss	18:50	0:00	/usr/lib/systemd/systemd-logind
root	178	0.0	0.3	1756096	13568	?	Ssl	18:50	0:00	/usr/libexec/wsl-pro-service -vv
root	180	0.0	0.0	3160	2048	hvc0	Ss+	18:50	0:00	/sbin/agetty -o -p -- \u --noclear --keep-baud - 115200,38400,9600 vt220
root	184	0.0	0.0	3116	1920	tty1	Ss+	18:50	0:00	/sbin/agetty -o -p -- \u --noclear - linux
syslog	185	0.0	0.1	222508	5632	?	Ssl	18:50	0:00	/usr/sbin/rsyslogd -n -iNONE
root	193	0.0	0.5	107012	22400	?	Ssl	18:50	0:00	/usr/bin/python3 /usr/share/unattended-upgrades/unattended-upgrade-shutdown --wai
root	300	0.0	0.0	3076	896	?	Ss	18:50	0:00	/init
root	301	0.0	0.0	3092	1024	?	S	18:50	0:00	-bash
user_ra+	302	0.0	0.1	6332	5376	pts/0	Ss	18:50	0:00	-bash
root	303	0.0	0.1	6664	4480	pts/1	Ss	18:50	0:00	/bin/login -f

- Gambar menunjukkan bahwa sistem Linux sedang dalam kondisi sangat stabil dan *idle*, dengan sumber daya memori dan CPU yang tersedia secara melimpah dan tidak ada beban kerja berat.

```
user_rafi@DESKTOP-AC7MIM2:~$ top
top - 20:56:39 up 2:06, 1 user, load average: 0.00, 0.00, 0.00
Tasks: 26 total, 1 running, 23 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.1 sy, 0.0 ni, 99.9 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 3787.2 total, 3198.6 free, 398.1 used, 271.7 buff/cache
MiB Swap: 1024.0 total, 1024.0 free, 0.0 used, 3389.2 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
169	message+	20	0	9688	5248	4608	S	0.3	0.1	0:00.46	dbus-daemon
1	root	20	0	21688	12288	9216	S	0.0	0.3	0:02.98	systemd
2	root	20	0	3072	1792	1792	S	0.0	0.0	0:00.02	init-systemd(Ub
7	root	20	0	3072	1792	1792	S	0.0	0.0	0:00.00	init
49	root	19	-1	66836	18884	17988	S	0.0	0.5	0:01.41	systemd-journal
102	root	20	0	25268	6272	4864	S	0.0	0.2	0:00.80	systemd-udevd
117	systemd+	20	0	21456	12416	10368	S	0.0	0.3	0:00.24	systemd-resolve
118	systemd+	20	0	91024	7552	6784	S	0.0	0.2	0:00.33	systemd-timesyn
168	root	20	0	4236	2560	2432	S	0.0	0.1	0:00.05	cron
176	root	20	0	17960	8576	7680	S	0.0	0.2	0:00.23	systemd-logind
178	root	20	0	1756096	13568	11008	S	0.0	0.3	0:00.44	wsl-pro-service
180	root	20	0	3160	2048	1920	S	0.0	0.1	0:00.01	agetty
184	root	20	0	3116	1792	1792	S	0.0	0.0	0:00.00	agetty
185	syslog	20	0	222508	5632	4480	S	0.0	0.1	0:00.22	rsyslogd
193	root	20	0	107012	22400	13184	S	0.0	0.6	0:00.32	unattended-upgr
300	root	20	0	3076	896	896	S	0.0	0.0	0:00.00	SessionLeader
301	root	20	0	3092	1024	896	S	0.0	0.0	0:00.47	Relay(302)
302	user_ra+	20	0	6332	5376	3584	S	0.0	0.1	0:00.50	bash
303	root	20	0	6664	4480	3840	S	0.0	0.1	0:00.01	login
348	user_ra+	20	0	20104	10880	9088	S	0.0	0.3	0:00.25	systemd
349	user_ra+	20	0	21156	3520	1792	S	0.0	0.1	0:00.00	(sd-pam)

- Gambar ini adalah memverifikasi *toolchain* pengembangan (gcc, bash), mencoba mengakses informasi *kernel*, dan yang terpenting, mengembangkan dan menjalankan program kustom untuk membaca data sistem dari *procfs*.

```

user_rafi@DESKTOP-AC7MIM2:~$ which bash
/usr/bin/bash
user_rafi@DESKTOP-AC7MIM2:~$ which gcc
/usr/bin/gcc
user_rafi@DESKTOP-AC7MIM2:~$ ls -lh /boot/vmlinuz-$(uname -r)
ls: cannot access '/boot/vmlinuz-6.6.87.2-microsoft-standard-WSL2': No such file or directory
user_rafi@DESKTOP-AC7MIM2:~$ cat /boot/config-$(uname -r) | grep "=y" | wc -l
cat: /boot/config-6.6.87.2-microsoft-standard-WSL2: No such file or directory
0
user_rafi@DESKTOP-AC7MIM2:~$ cat /boot/config-$(uname -r) | grep "=m" | wc -l
cat: /boot/config-6.6.87.2-microsoft-standard-WSL2: No such file or directory
0
user_rafi@DESKTOP-AC7MIM2:~$ lsmod | wc -l
19
user_rafi@DESKTOP-AC7MIM2:~$ nano procfs_read.c
user_rafi@DESKTOP-AC7MIM2:~$ gcc -o procfs_read procfs_read.c
procfs_read.c: In function 'main':
procfs_read.c:32:60: warning: implicit declaration of function 'getpid' [-Wimplicit-function-declaration]
   32 |     snprintf(proc_path, sizeof(proc_path), "/proc/%d/status", getpid());
      |                                     ^~~~~~
user_rafi@DESKTOP-AC7MIM2:~$ ./procfs_read
=== CPU Information ===
processor       : 0
vendor_id      : GenuineIntel
cpu family     : 6
model          : 140
model name     : 11th Gen Intel(R) Core(TM) i3-1115G4 @ 3.00GHz
stepping       : 1
microcode      : 0xffffffff
cpu MHz        : 2188.799
cache size     : 6144 KB
physical id    : 0

```

1.9 Building and Booting OS

1.9.1 Analisis Boot Process

1.9.2 Kernel Parameters

1.9.3 Simple Bootloader (Konsep)

- Gambar ini adalah melakukan diagnostik dan *troubleshooting* kecepatan *boot* sistem dengan mencari tahu komponen mana yang menghabiskan waktu paling banyak.

```
user_rafi@DESKTOP-AC7MIM2:~$ dmesg | less
[2]+  Stopped                  dmesg | less
user_rafi@DESKTOP-AC7MIM2:~$ systemd-analyze
Startup finished in 5.557s (userspace)
graphical.target reached after 5.531s in userspace.
user_rafi@DESKTOP-AC7MIM2:~$ systemd-analyze blame
25.543s motd-news.service
 2.338s landscape-client.service
 2.085s dev-sdd.device
 1.236s snapd.seeded.service
 1.052s snapd.service
 791ms wsl-pro.service
 500ms systemd-udev.service
 489ms systemd-resolved.service
 403ms systemd-udev-trigger.service
 331ms keyboard-setup.service
 324ms systemd-journald.service
 266ms user@1000.service
 265ms systemd-sysctl.service
 249ms systemd-tmpfiles-setup.service
 217ms polkit.service
```

- Gambar ini berfokus pada diagnostik waktu *boot* dan sumber daya sistem (systemd-analyze) dan upaya untuk menyelidiki dan memodifikasi komponen *boot* tingkat rendah (kernel command line, bootloader). walaupun banyak perintah pemeriksaan *bootloader* tradisional gagal karena sifat lingkungan WSL

```

user_rafi@DESKTOP-AC7MIM2:~$ systemd-analyze critical-chain
The time when unit became active or started is printed after the "@" character.
The time the unit took to start is printed after the "+" character.

graphical.target @5.531s
└─multi-user.target @5.530s
   └─snapd.seeded.service @3.207s +1.236s
      └─basic.target @3.173s
         └─sockets.target @3.173s
            └─snapd.socket @3.098s +74ms
               └─sysinit.target @3.086s
                  └─systemd-resolved.service @2.596s +489ms
                     └─systemd-tmpfiles-setup.service @2.253s +249ms
                        └─systemd-journal-flush.service @2.076s +158ms
                           └─systemd-journald.service @1.587s +324ms
                              └─systemd-journald.socket @1.550s
                                 └─.mount @766ms
                                    └─.slice @759ms

user_rafi@DESKTOP-AC7MIM2:~$ cat /boot/grub/grub.cfg | head -50
cat: /boot/grub/grub.cfg: No such file or directory
user_rafi@DESKTOP-AC7MIM2:~$ lsinitramfs /boot/initrd.img-$(uname -r) | head -20
Command 'lsinitramfs' not found, but can be installed with:
sudo apt install initramfs-tools-core
user_rafi@DESKTOP-AC7MIM2:~$ cat /proc/cmdline
initrd=\initrd.img WSL_ROOT_INIT=1 panic=-1 nr_cpus=4 hv_utils.timesync_implicit=1 console=hvc0 debug ptty.legacy_count=0 WSL_ENABLE_CRASH_DUMP=1
user_rafi@DESKTOP-AC7MIM2:~$ sudo nano /etc/default/grub
[sudo] password for user_rafi:
user_rafi@DESKTOP-AC7MIM2:~$ nano bootloader_concept.asm

```

1.10 OS Debugging

1.10.1 Using strace

- Inti dari kode ini adalah untuk menganalisis langkah demi langkah bagaimana sebuah program sederhana (ls) dimuat dan dijalankan oleh sistem operasi, mulai dari inisiasi proses, pemuatan pustaka, hingga alokasi memori, yang semuanya diungkapkan melalui panggilan sistem strace.

```
user_rafi@DESKTOP-AC7MIM x + v
user_rafi@DESKTOP-AC7MIM2:~$ strace -v ls
execve("/usr/bin/ls", ["ls"], ["SHELL=/bin/bash", "WSL2_GUI_APPS_ENAB
home/user_rafi", "LOGNAME=user_rafi", "HOME=/home/user_rafi", "LANG=C
01;34:ln=01;36"... , "WAYLAND_DISPLAY=wayland-0", "LESSCLOSE=/usr/bin/
esspipe %s", "USER=user_rafi", "DISPLAY=:0", "SHLVL=1", "XDG_RUNTIME_
LENV=", "XDG_DATA_DIRS=/usr/local/share:/"... , "PATH=/usr/local/sbin:
E=x86_64", "PULSE_SERVER=unix:/mnt/wslg/Puls"... , "_=/usr/bin/strace"
brk(NULL) = 0x5d04b3b29000
mmap(NULL, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1,
access("/etc/ld.so.preload", R_OK) = -1 ENOENT (No such file or
openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_dev=makedev(0x8, 0x30), st_ino=2595, st_mode=S_IFREG|0644
, st_size=19879, st_atime=1760276478 /* 2025-10-12T20:41:18.283136784
0-12T20:41:18.039136776+0700 */ , st_mtime_nsec=39136776, st_ctime=1760
51136776}) = 0
mmap(NULL, 19879, PROT_READ, MAP_PRIVATE, 3, 0) = 0x76b1d5f59000
close(3) = 0
```

- Inti dari output ini adalah untuk mengukur dan mengaudit kinerja program ls dengan tepat. Ia mengungkapkan bahwa sebagian besar waktu eksekusi program dihabiskan untuk operasi input/output file dasar (openat, close, fstat), bukan pada komputasi, yang merupakan karakteristik umum dari utilitas sistem seperti ls.

```

user_rafi@DESKTOP-AC7MIM x + v
user_rafi@DESKTOP-AC7MIM2:~$ strace -c ls
amiruddin      amiruddin5      hello_module.c  math_ops.h      myshell
amiruddin.c    amiruddin5.c    libmathops.so   math_ops.i      myshell.c
amiruddin2     bootloader_concept.asm log.txt         math_ops.o      os_specific.c
amiruddin2.c   buggy           main.c          math_ops.s      output.txt
amiruddin3     buggy.c         main.i          memleak         proc_monitor
amiruddin3.c   file_monitor    main.o          memleak.c       proc_monitor.c
amiruddin4     file_monitor.c  main.s          my_daemon       process_manager
amiruddin4.c   hasil.txt       math_ops.c      my_daemon.c     process_manager.c
% time        seconds  usecs/call      calls      errors  syscall
-----
27.25         0.000972      27              35          13  openat
12.78         0.000456      19              24           0   close
11.89         0.000424      18              23           0   fstat
11.24         0.000401      13              30           0   mmap
9.81          0.000350      35              10           0   read
9.48          0.000338      42              8            0   write
4.96          0.000177      35              5            0   mprotect
2.13          0.000076      38              2            0   getdents64
2.02          0.000072      36              2            1   statfs
1.29          0.000046      46              1            0   munmap
1.26          0.000045      15              3            0   brk
1.18          0.000042      21              2            0   ioctl

```

- Gambar dibawah ini menjelaskan perintah ini bertujuan untuk menganalisis struktur dan perilaku internal program ls dengan merekam semua permintaan yang dibuatnya ke kernel. Output yang ditampilkan adalah bukti rinci bagaimana program dimuat, mengalokasikan memori, dan memuat pustaka yang dibutuhkan sebelum dapat menjalankan fungsi utamanya (mencantumkan file).

```

user_rafi@DESKTOP-AC7MIM  X  +  v
user_rafi@DESKTOP-AC7MIM2:~$ strace -o trace.txt ls
amiruddin      amiruddin5      hello_module.c  math_ops.h      myshell
amiruddin.c    amiruddin5.c    libmathops.so   math_ops.i       myshell.c
amiruddin2     bootloader_concept.asm  log.txt         math_ops.o       os_specific.c
amiruddin2.c   buggy           main.c          math_ops.s       output.txt
amiruddin3     buggy.c         main.i          memleak          proc_monitor
amiruddin3.c   file_monitor    main.o          memleak.c        proc_monitor.c
amiruddin4     file_monitor.c  main.s          my_daemon        process_manager
amiruddin4.c   hasil.txt       math_ops.c      my_daemon.c      process_manager.c
user_rafi@DESKTOP-AC7MIM2:~$ cat trace.txt
execve("/usr/bin/ls", ["ls"], 0x7ffcd3fe4e50 /* 26 vars */) = 0
brk(NULL)                               = 0x592560557000
mmap(NULL, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7a3f558d0000
access("/etc/ld.so.preload", R_OK)      = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=19879, ...}) = 0
mmap(NULL, 19879, PROT_READ, MAP_PRIVATE, 3, 0) = 0x7a3f558d0000
close(3)                                = 0
openat(AT_FDCWD, "/lib/x86_64-linux-gnu/libselinux.so.1", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0..."..., 832) =
fstat(3, {st_mode=S_IFREG|0644, st_size=174472, ...}) = 0
mmap(NULL, 181960, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x7a3f558a3000
mmap(0x7a3f558a9000, 118784, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0) = 0x7a3f558a9000
mmap(0x7a3f558c6000, 24576, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0) = 0x7a3f558c6000
mmap(0x7a3f558cc000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0) = 0x7a3f558cc000
mmap(0x7a3f558ce000, 5832, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0x7a3f558ce000
close(3)                                = 0

```

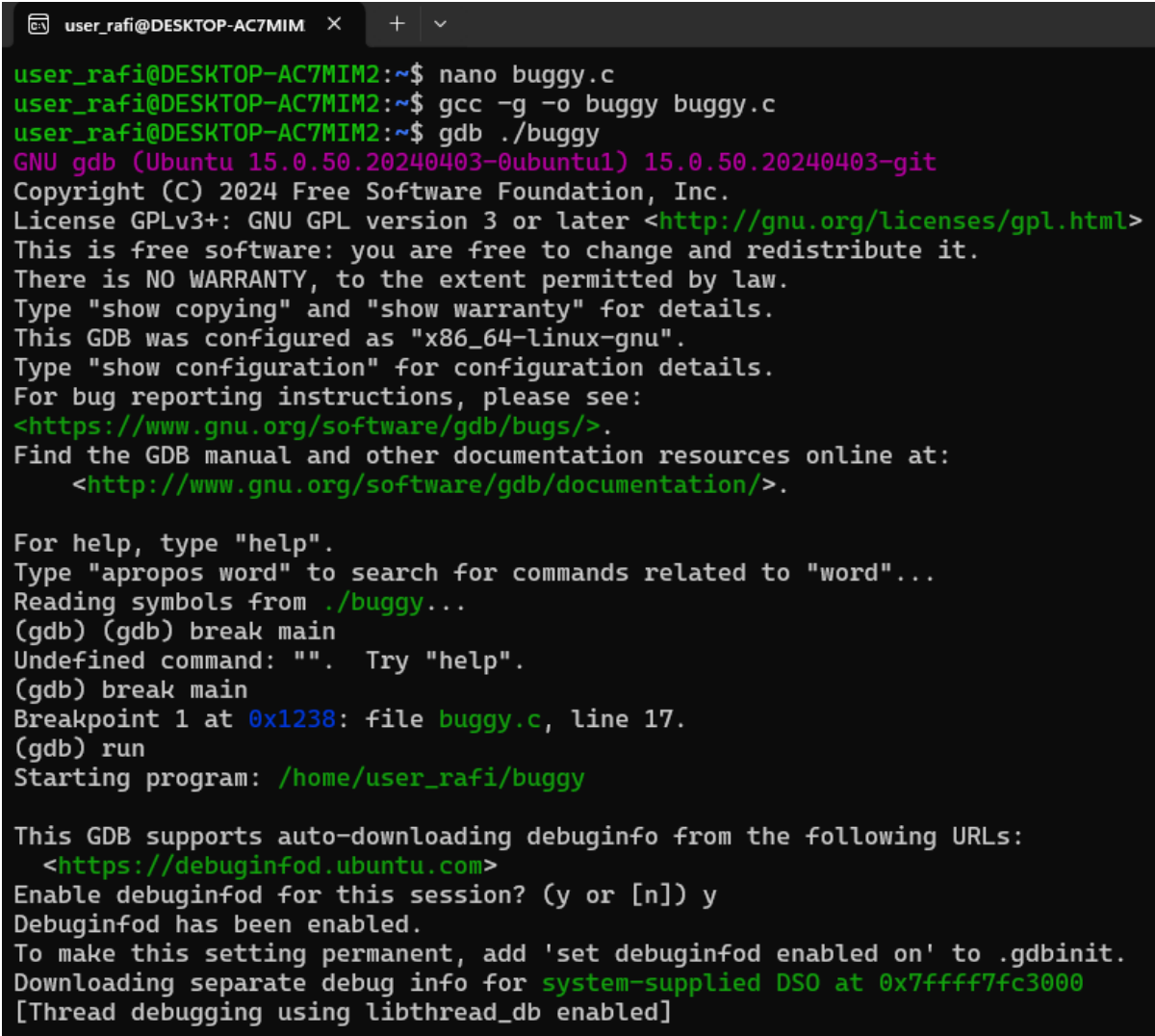
1.10.2 Using ltrace

- perintah ini menunjukkan upaya untuk memasang alat diagnostik (ltrace) dan menggunakannya untuk menganalisis perilaku program userspace (ls) pada tingkat fungsi pustaka, memberikan wawasan tentang bagaimana program berinteraksi dengan pustaka C dan lingkungan lokalnya.

```
user_rafi@DESKTOP-AC7MIM:~$ sudo apt install ltrace
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
  ltrace
0 upgraded, 1 newly installed, 0 to remove and 38 not upgraded.
Need to get 134 kB of archives.
After this operation, 391 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble/universe amd64 ltrace amd64 0.7.3-6.4ubuntu3 [134 kB]
Fetched 134 kB in 2s (78.3 kB/s)
Selecting previously unselected package ltrace.
(Reading database ... 46403 files and directories currently installed.)
Preparing to unpack .../ltrace_0.7.3-6.4ubuntu3_amd64.deb ...
Unpacking ltrace (0.7.3-6.4ubuntu3) ...
Setting up ltrace (0.7.3-6.4ubuntu3) ...
Processing triggers for man-db (2.12.0-4build2) ...
user_rafi@DESKTOP-AC7MIM:~$ ltrace ls
strchr("ls", '/') = nil
setlocale(LC_ALL, "") = "C.UTF-8"
bindtextdomain("coreutils", "/usr/share/locale") = "/usr/share/locale"
textdomain("coreutils") = "coreutils"
__cxa_atexit(0x556b8dcc3880, 0, 0x556b8dcd0008, 0) = 0
getopt_long(1, 0x7ffd512f4a48, "abdcdfghiklmnopqrstuvw:ABCDEFGH", -1) = -1
getenv("LS_BLOCK_SIZE") = nil
getenv("BLOCK_SIZE") = nil
getenv("BLOCKSIZE") = nil
getenv("POSIXLY_CORRECT") = nil
getenv("BLOCK_SIZE") = nil
isatty(1) = 1
ioctl(1, 21523, 0x7ffd512f4860) = 0
getenv("TABSIZE") = nil
getenv("QUOTING_STYLE") = nil
```

1.10.3 Using GDB

- Gambar ini mendemonstrasikan alur kerja standar seorang developer: menulis kode (nano), mengompilasinya untuk debugging (gcc -g), dan kemudian menggunakan alat profesional (gdb) untuk menelusuri eksekusi program langkah demi langkah untuk mengisolasi dan memperbaiki bug.



```
user_rafi@DESKTOP-AC7MIM x + v
user_rafi@DESKTOP-AC7MIM2:~$ nano buggy.c
user_rafi@DESKTOP-AC7MIM2:~$ gcc -g -o buggy buggy.c
user_rafi@DESKTOP-AC7MIM2:~$ gdb ./buggy
GNU gdb (Ubuntu 15.0.50.20240403-0ubuntu1) 15.0.50.20240403-git
Copyright (C) 2024 Free Software Foundation, Inc.
License GPLv3+: GNU GPL version 3 or later <http://gnu.org/licenses/gpl.html>
This is free software: you are free to change and redistribute it.
There is NO WARRANTY, to the extent permitted by law.
Type "show copying" and "show warranty" for details.
This GDB was configured as "x86_64-linux-gnu".
Type "show configuration" for configuration details.
For bug reporting instructions, please see:
<https://www.gnu.org/software/gdb/bugs/>.
Find the GDB manual and other documentation resources online at:
<http://www.gnu.org/software/gdb/documentation/>.

For help, type "help".
Type "apropos word" to search for commands related to "word"...
Reading symbols from ./buggy...
(gdb) (gdb) break main
Undefined command: "". Try "help".
(gdb) break main
Breakpoint 1 at 0x1238: file buggy.c, line 17.
(gdb) run
Starting program: /home/user_rafi/buggy

This GDB supports auto-downloading debuginfo from the following URLs:
<https://debuginfod.ubuntu.com>
Enable debuginfod for this session? (y or [n]) y
Debuginfod has been enabled.
To make this setting permanent, add 'set debuginfod enabled on' to .gdbinit.
Downloading separate debug info for system-supplied DSO at 0x7ffff7fc3000
[Thread debugging using libthread_db enabled]
```

- Gambar ini adalah menelusuri dampak dari pemanggilan fungsi kunci (`buggy_function(5)`), memverifikasi bahwa program berjalan dengan sukses hingga akhir, dan mengakhiri sesi *debugging* setelah program selesai dieksekusi.

```

user_rafi@DESKTOP-AC7MIM X + v

Breakpoint 1, main () at buggy.c:17
17      buggy_function(5);
(gdb) next
array[0] = 0
array[1] = 2
array[2] = 4
array[3] = 6
array[4] = 8
18      return 0;
(gdb) step
19      }
(gdb) print n
No symbol "n" in current context.
(gdb) continue
Continuing.
[Inferior 1 (process 787) exited normally]
(gdb) backtrace
No stack.
(gdb) quit

```

1.10.4 Kernel Debugging dengan dmesg

- `dmesg` memberikan log historis kernel yang membuktikan bahwa sistem Linux ini adalah lingkungan WSL2 yang diinisialisasi sebagai mesin virtual oleh Microsoft Hyper-V, dengan detail mengenai alokasi sumber daya dan integrasi kernel dengan hypervisor.

```

user_rafi@DESKTOP-AC7MIM2:~$ dmesg
[ 0.000000] Linux version 6.6.87.2-microsoft-standard-WSL2 (root@439a258ad544) (gcc (GCC) 11.2.0, GNU ld (GNU Binutils) 2.37) #1 SMP PR
MIC Thu Jun  5 18:30:46 UTC 2025
[ 0.000000] Command line: initrd=\initrd.img WSL_ROOT_INIT=1 panic=-1 nr_cpus=4 hv_utils.timesync_implicit=1 console=hvc0 debug pty.leg
0 WSL_ENABLE_CRASH_DUMP=1
[ 0.000000] KERNEL supported cpus:
[ 0.000000]   Intel GenuineIntel
[ 0.000000]   AMD AuthenticAMD
[ 0.000000] BIOS-provided physical RAM map:
[ 0.000000] BIOS-e820: [mem 0x0000000000000000-0x0000000000009fff] usable
[ 0.000000] BIOS-e820: [mem 0x000000000000e000-0x000000000000e0fff] reserved
[ 0.000000] BIOS-e820: [mem 0x00000000000100000-0x000000000001ffff] ACPI data
[ 0.000000] BIOS-e820: [mem 0x00000000000200000-0x000000000007bffff] usable
[ 0.000000] NX (Execute Disable) protection: active
[ 0.000000] APIC: Static calls initialized
[ 0.000000] DMI not present or invalid.
[ 0.000000] Hypervisor detected: Microsoft Hyper-V
[ 0.000000] Hyper-V: privilege flags low 0xae7f, high 0x3b8030, hints 0xa4e24, misc 0xe4bed7b6
[ 0.000000] Hyper-V: Nested features: 0x3e0101
[ 0.000000] Hyper-V: LAPIC Timer Frequency: 0xc3500
[ 0.000000] Hyper-V: Using hypercall for remote TLB flush
[ 0.000000] clocksource: hyperv_clocksource_tsc_page: mask: 0xffffffffffffffff max_cycles: 0x24e6a1710, max_idle_ns: 440795202120 ns
[ 0.000000] clocksource: hyperv_clocksource_msr: mask: 0xffffffffffffffff max_cycles: 0x24e6a1710, max_idle_ns: 440795202120 ns
[ 0.000000] tsc: Detected 2188.700 MHz processor

```

- Perintah ini bertujuan untuk memantau log sistem *kernel* secara dinamis, dan *output* yang ditampilkan menguraikan konfigurasi *boot* sistem dan dengan jelas mengidentifikasi bahwa sistem Linux tersebut beroperasi di lingkungan virtual WSL2/Hyper-V.

```
user_rafi@DESKTOP-AC7MIM2:~$ dmesg -w
[ 0.000000] Linux version 6.6.87.2-microsoft-standard-WSL2 (root@439a258ad544) (gcc (GCC) 11.2.0, GNU ld (GNU Binutils) 2.37) #1 SMP PRE
MIC Thu Jun 5 18:30:46 UTC 2025
[ 0.000000] Command line: initrd=\initrd.img WSL_ROOT_INIT=1 panic=-1 nr_cpus=4 hv_utils.timesync_implicit=1 console=hvc0 debug pty.lega
0 WSL_ENABLE_CRASH_DUMP=1
[ 0.000000] KERNEL supported cpus:
[ 0.000000] Intel GenuineIntel
[ 0.000000] AMD AuthenticAMD
[ 0.000000] BIOS-provided physical RAM map:
[ 0.000000] BIOS-e820: [mem 0x0000000000000000-0x0000000000000fff] usable
[ 0.000000] BIOS-e820: [mem 0x0000000000000000-0x0000000000000fff] reserved
[ 0.000000] BIOS-e820: [mem 0x0000000000000000-0x0000000000001ffff] ACPI data
[ 0.000000] BIOS-e820: [mem 0x0000000000000000-0x0000000000000fff] usable
[ 0.000000] NX (Execute Disable) protection: active
[ 0.000000] APIC: Static calls initialized
[ 0.000000] DMI not present or invalid.
[ 0.000000] Hypervisor detected: Microsoft Hyper-V
[ 0.000000] Hyper-V: privilege flags low 0xae7f, high 0x3b8030, hints 0xa4e24, misc 0xe4bed7b6
[ 0.000000] Hyper-V: Nested features: 0x3e0101
[ 0.000000] Hyper-V: LAPIC Timer Frequency: 0xc3500
[ 0.000000] Hyper-V: Using hypercall for remote TLB flush
[ 0.000000] clocksource: hyperv_clocksource_tsc_page: mask: 0xffffffffffffffff max_cycles: 0x24e6a1710, max_idle_ns: 440795202120 ns
[ 0.000000] clocksource: hyperv_clocksource_msr: mask: 0xffffffffffffffff max_cycles: 0x24e6a1710, max_idle_ns: 440795202120 ns
[ 0.000000] tsc: Detected 2188.799 MHz processor
[ 0.001011] e820: update [mem 0x00000000-0x00000fff] usable ==> reserved
[ 0.001061] e820: remove [mem 0x00000000-0x00000fff] usable
```

- Inti dari perintah ini adalah untuk mendiagnosis kesehatan dan *bug* sistem. Hasilnya menunjukkan bahwa meskipun sistem berjalan, ia mengalami berbagai masalah yang berhubungan dengan virtualisasi (Hyper-V/WSL), *driver* grafis, dan kinerja jaringan TCP karena keterbatasan atau *bug* di lingkungan virtual tersebut.

```
user_rafi@DESKTOP-AC7MIM2:~$ dmesg -l err
[ 0.188720] PCI: Fatal: No config space access function found
[ 4.039997] misc dxg: dxgk: dxgkio_is_feature_enabled: Ioctl failed: -22
[ 4.134124] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -22
[ 4.135252] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -22
[ 4.151203] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -22
[ 4.153881] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -2
[ 7.447731] WSL (190) ERROR: CheckConnection: connect() failed: 101
[ 2770.545077] WSL (190) ERROR: CheckConnection: connect() failed: 101
user_rafi@DESKTOP-AC7MIM2:~$ dmesg -l warn
[ 0.115879] Hyper-V: Disabling IBT because of Hyper-V bug
[ 0.225238] PCI: System does not support PCI
[ 0.361354] device-mapper: core: CONFIG_IMA_DISABLE_HTABLE is disabled. Duplicate IMA measurements w
d in the IMA log.
[ 3.296202] pulseaudio[224]: memfd_create() called without MFD_EXEC or MFD_NOEXEC_SEAL set
[ 3.587909] Failed to connect to bus: No such file or directory
[ 3.943470] Failed to connect to bus: No such file or directory
[ 5.415600] systemd-journald[49]: File /var/log/journal/a92c99e25adc49c1a1547072047067cb/system.journ
ncleanly shut down, renaming and replacing.
[ 12.264870] TCP: eth0: Driver has suspect GRO implementation, TCP performance may be compromised.
```

1.10.5 Memory Leak Detection dengan Valgrind

- Rangkaian aktivitas ini menunjukkan langkah-langkah optimasi dan diagnostik sistem—pengguna pertama menyiapkan alat untuk memperbaiki masalah kode program (Valgrind), dan kedua, menganalisis masalah kecepatan sistem (systemd-analyze blame).

```
user_rafi@DESKTOP-AC7MIM2:~$ sudo apt install valgrind
[sudo] password for user_rafi:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  gdb libbabeltrace1 libc6-dbg libdebuginfod-common libdebuginfod1t64 libipt2 libsource-highlight-common
  libsource-highlight4t64
Suggested packages:
  gdb-doc gdbserver valgrind-dbg valgrind-mpi kcachegrind alleyoop valkyrie
The following NEW packages will be installed:
  gdb libbabeltrace1 libc6-dbg libdebuginfod-common libdebuginfod1t64 libipt2 libsource-highlight-common
  libsource-highlight4t64 valgrind
0 upgraded, 9 newly installed, 0 to remove and 38 not upgraded.
Need to get 26.9 MB of archives.
After this operation, 105 MB of additional disk space will be used.
Do you want to continue? [Y/n] y
```

- terminal ini menunjukkan pendekatan dua arah: Administrasi Sistem (mengoptimalkan *boot* sistem dan memeriksa *hardware*) dan Pengembangan Perangkat Lunak (membuat dan memperbaiki masalah *memory leak* pada kode C).

```
user_rafi@DESKTOP-AC7MIM2:~$ cat > memleak.c << 'EOF'
> ^C
user_rafi@DESKTOP-AC7MIM2:~$ cat > memleak.c << 'EOF'
#include <stdlib.h>
int main() {
    int *ptr = (int*)malloc(100 * sizeof(int));
    // Forgot to free(ptr)
    return 0;
}
EOF
user_rafi@DESKTOP-AC7MIM2:~$ gcc -g -o memleak memleak.c
user_rafi@DESKTOP-AC7MIM2:~$ valgrind --leak-check=full ./memleak
==2781== Memcheck, a memory error detector
==2781== Copyright (C) 2002-2022, and GNU GPL'd, by Julian Seward et al.
==2781== Using Valgrind-3.22.0 and LibVEX; rerun with -h for copyright info
==2781== Command: ./memleak
==2781==
==2781==
==2781== HEAP SUMMARY:
==2781==    in use at exit: 400 bytes in 1 blocks
==2781==   total heap usage: 1 allocs, 0 frees, 400 bytes allocated
==2781==
==2781== 400 bytes in 1 blocks are definitely lost in loss record 1 of 1
==2781==    at 0x4846828: malloc (in /usr/libexec/valgrind/vgpreload_memcheck-amd64-linux.so)
==2781==    by 0x10915E: main (memleak.c:3)
```

1.10.6 Performance Analysis

- kode ini adalah dokumentasi dari seorang administrator atau developer yang sedang menyiapkan lingkungan dan melakukan audit kinerja dasar pada instalasi Linux-nya.

```
user_rafi@DESKTOP-AC7MIM2:~$ sudo apt install linux-tools-common linux-tools-generic
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  hwdata libnuma1 libpci3 libtraceevent1 linux-tools-6.8.0-85 linux-tools-6.8.0-85-generic
The following NEW packages will be installed:
  hwdata libnuma1 libpci3 libtraceevent1 linux-tools-6.8.0-85 linux-tools-6.8.0-85-generic linux-tools-common
  linux-tools-generic
0 upgraded, 8 newly installed, 0 to remove and 38 not upgraded.
Need to get 6340 kB of archives.
After this operation, 21.0 MB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://archive.ubuntu.com/ubuntu noble/main amd64 libnuma1 amd64 2.0.18-1build1 [23.3 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble/main amd64 libpci3 amd64 1:3.10.0-2build1 [36.5 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libtraceevent1 amd64 1:1.8.2-1ubuntu2.1 [55.3 kB]
Get:4 http://archive.ubuntu.com/ubuntu noble/main amd64 hwdata all 0.379-1 [29.1 kB]
Get:5 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 linux-tools-common all 6.8.0-85.85 [762 kB]
Get:6 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 linux-tools-6.8.0-85 amd64 6.8.0-85.85 [5421 kB]
Get:7 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 linux-tools-6.8.0-85-generic amd64 6.8.0-85.85 [1804 B]
Get:8 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 linux-tools-generic amd64 6.8.0-85.85 [10.8 kB]
Fetched 6340 kB in 5s (1225 kB/s)
```

- Gambar ini menunjukkan alur kerja diagnostik lengkap: mulai dari pemeriksaan status dasar dan hardware, identifikasi bottleneck pada startup sistem, hingga upaya instalasi alat profiling tingkat lanjut untuk debugging kinerja yang lebih spesifik.

```
user_rafi@DESKTOP-AC7MIM2:~$ sudo perf trace ls
WARNING: perf not found for kernel 6.6.87.2-microsoft

You may need to install the following packages for this specific kernel:
  linux-tools-6.6.87.2-microsoft-standard-WSL2
  linux-cloud-tools-6.6.87.2-microsoft-standard-WSL2

You may also want to install one of the following packages to keep up to date:
  linux-tools-standard-WSL2
  linux-cloud-tools-standard-WSL2
user_rafi@DESKTOP-AC7MIM2:~$ sudo perf top
WARNING: perf not found for kernel 6.6.87.2-microsoft

You may need to install the following packages for this specific kernel:
  linux-tools-6.6.87.2-microsoft-standard-WSL2
  linux-cloud-tools-6.6.87.2-microsoft-standard-WSL2

You may also want to install one of the following packages to keep up to date:
  linux-tools-standard-WSL2
  linux-cloud-tools-standard-WSL2
```